

COSE471: Data Science

Understanding and Preprocessing Data

Department of Computer Science and Engineering

Korea University

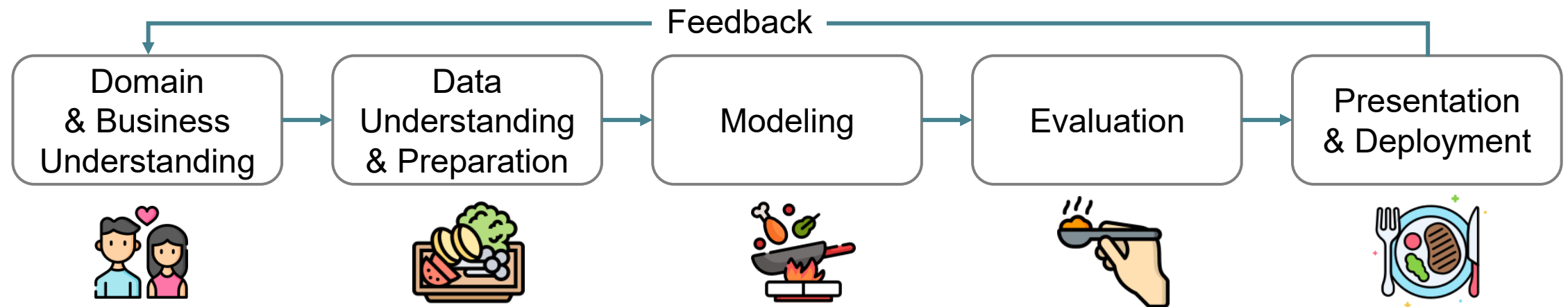
Susik Yoon

March 13, 2026

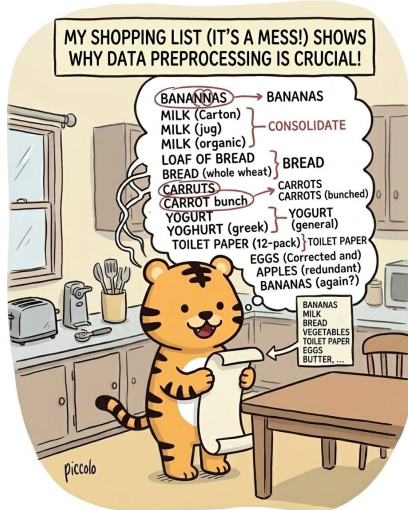
RECAP

❑ Data science

- Comparison with artificial intelligence
- Definition: the process of discovering interesting patterns and knowledge from large amounts of data
- Standard process
 - Domain/business understanding → Data understanding/preprocessing → Modeling → Evaluation → Presentation/Deployment



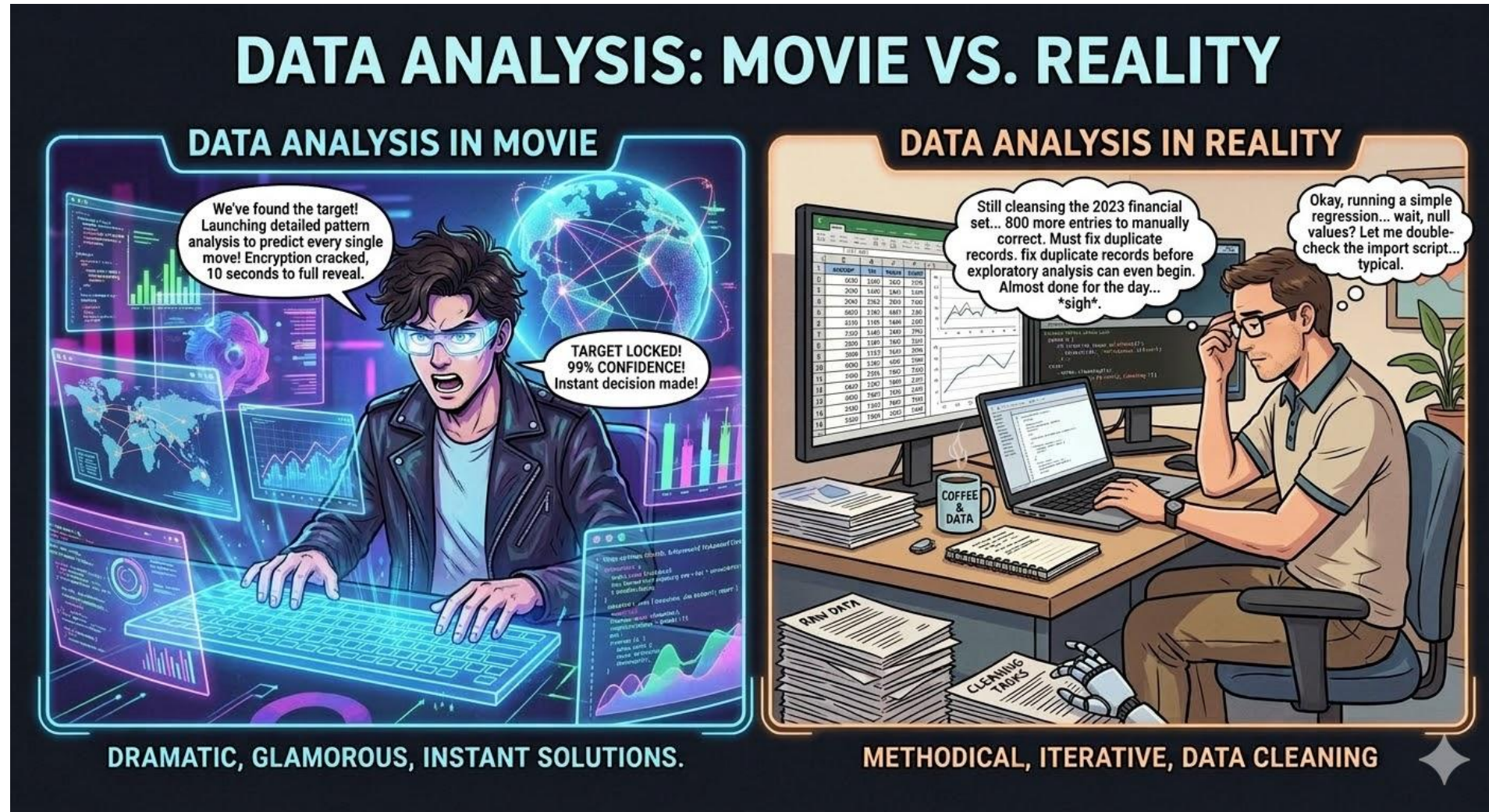
RECAP



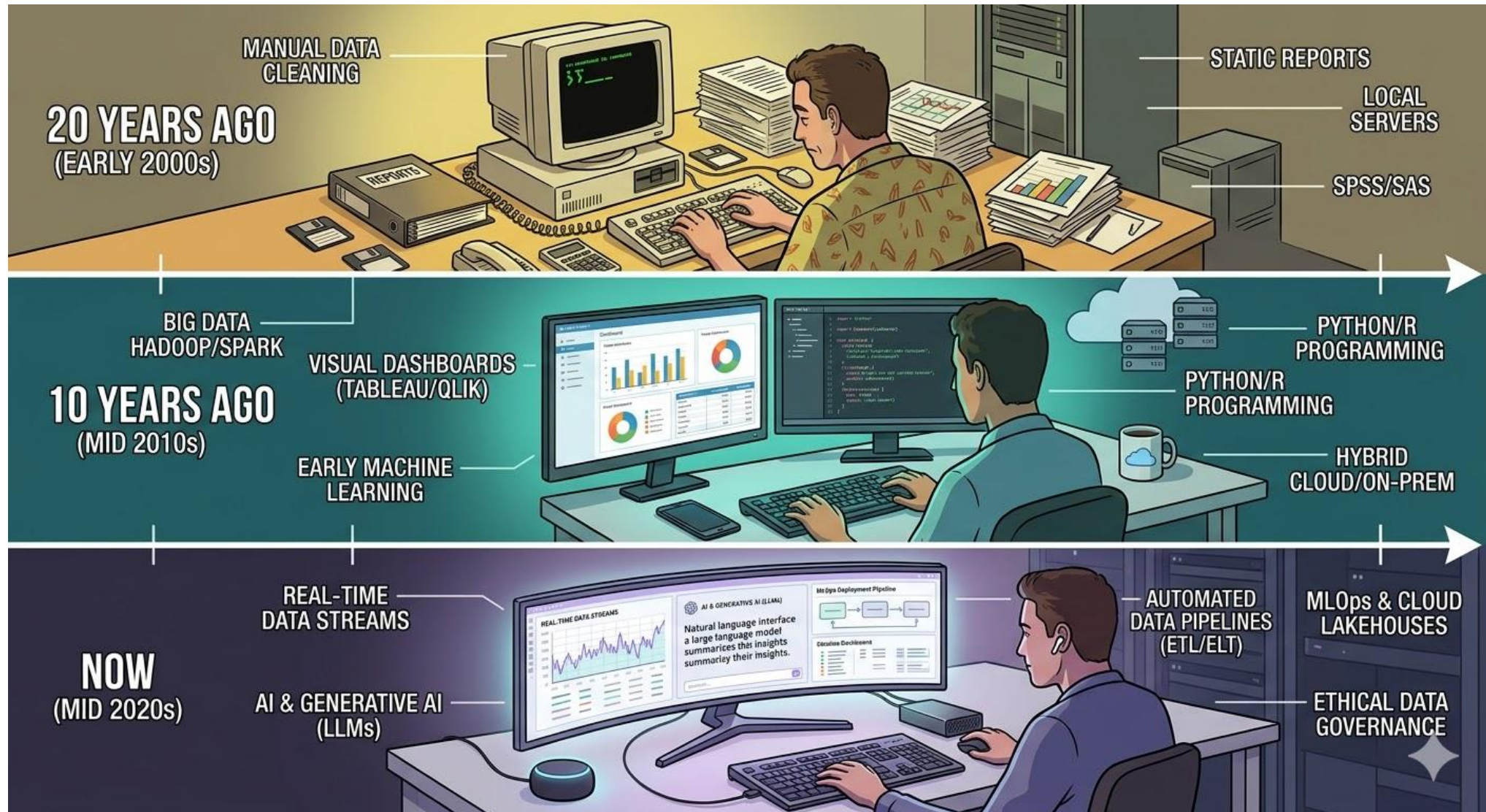
Course Schedule

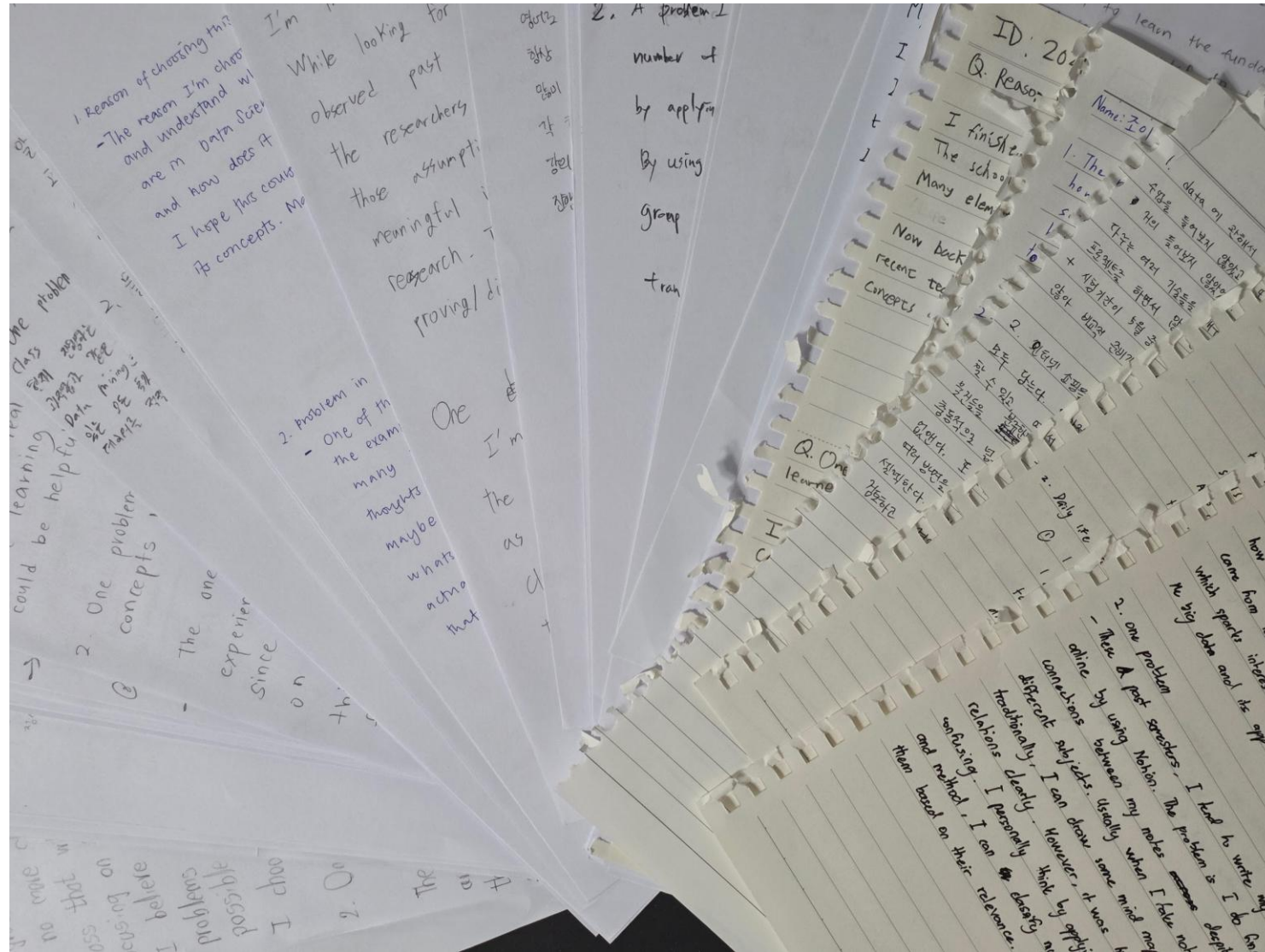
Week	Day	Content	HW/Project	Note
1	Mar 06	Introduction		
2	Mar 13	Data		
3	Mar 20	Pattern mining		
4	Mar 27	Tutorial 1	● HW1	<i>Latex , Preprocessing/Pattern mining</i>
5	Apr 03	Clustering	●	
6	Apr 10	Outlier Detection & Invited Talk	●	<i>An invited talk from Meta</i>
7	Apr 17	Tutorial 2	● HW2	<i>Clustering, Outlier detection</i>
8	Apr 24	Classification	●	
9	May 01	Classification	●	
10	May 08	Tutorial 3	●	<i>Classification, Feature engineering</i>
11	May 15	Mid-term exam		<i>Offline, Closed-book, No electronic devices</i>
12	May 22	Review & Project Notice	● Project	
13	May 29	Project discussion	●	
14	Jun 05	Project discussion	●	
15	Jun 12	Final presentation	●	
16	Jun 19	Wrap-up and Final report	●	

Data Analysis in Movie vs. in Reality

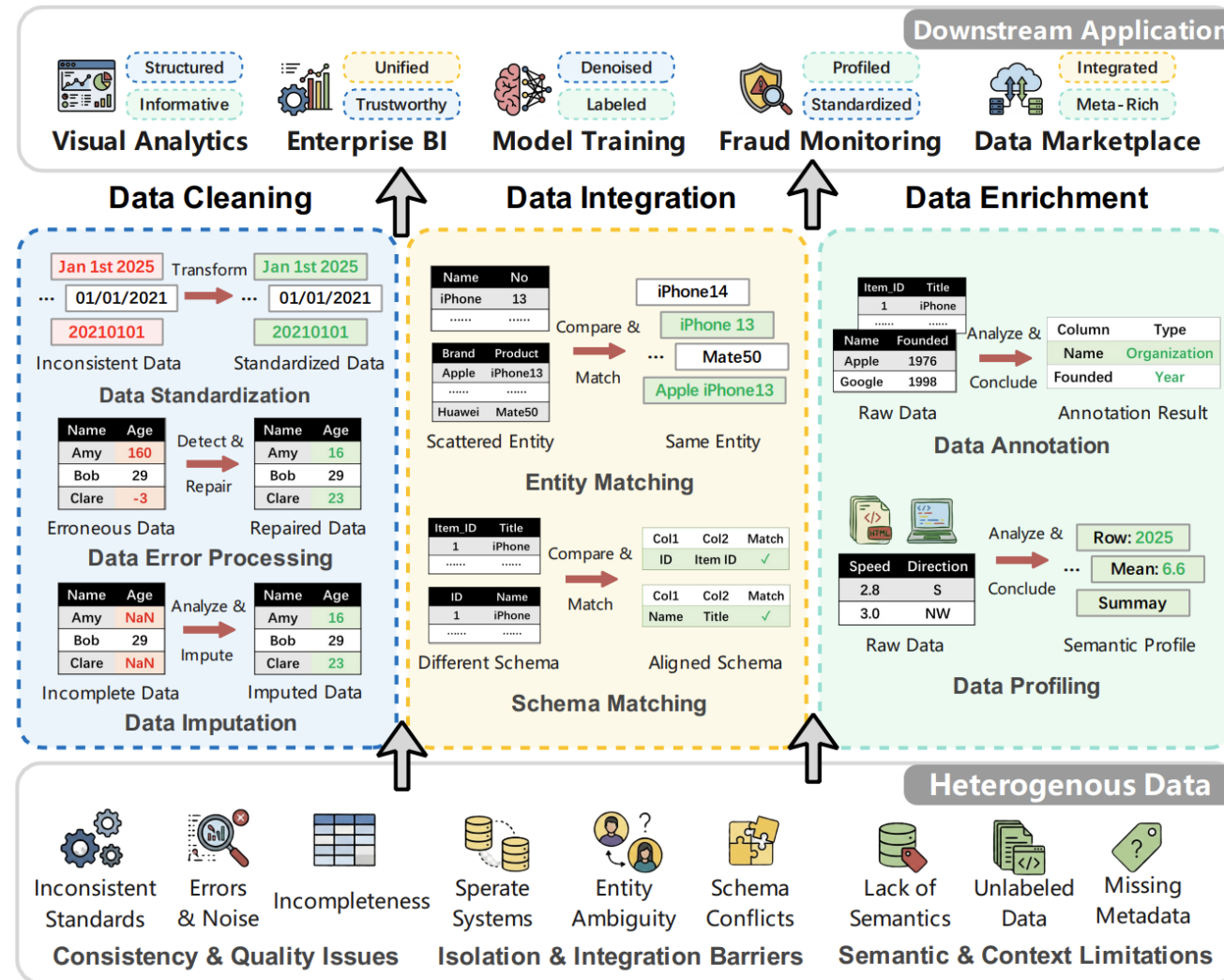


Data Analysis in 2000s, 2010s, and now





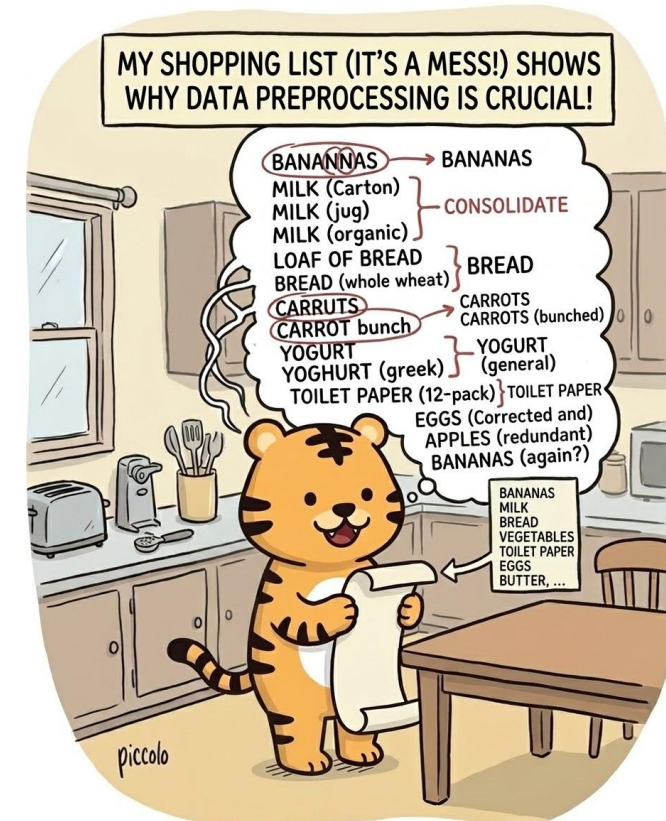
Data Preparation for Real-World Applications



<https://arxiv.org/pdf/2601.17058>

Outline

- ❑ Diversity of data
- ❑ Understanding Data
- ❑ Preprocessing Data



Diversity of Data

□ Data structures

- Structured data
 - Uniform, table-like structures, defined by data dictionaries, with a fixed set of attributes, each with a fixed set of value ranges and semantic meaning
 - e.g., relational database, data cube, data matrix
- Unstructured data
 - Text data and multimedia data
 - e.g., document, audio, image, video
- Semi-structured data
 - Data objects containing a set value, a small set of heterogeneous typed values, or nested structures, with flexible and dynamic structures
 - e.g., XML, graph, network

Structured data

ID	Name	Age	Degree
1	John	18	B.Sc.
2	David	31	Ph.D.
3	Robert	51	Ph.D.
4	Rick	26	M.Sc.
5	Michael	19	B.Sc.

Unstructured data

The university has 5600 students.
 John's ID is number 1, he is 18 years old and already holds a B.Sc. degree.
 David's ID is number 2, he is 31 years old and holds a Ph.D. degree. Robert's ID is number 3, he is 51 years old and also holds the same degree as David, a Ph.D. degree.

Semi-structured data

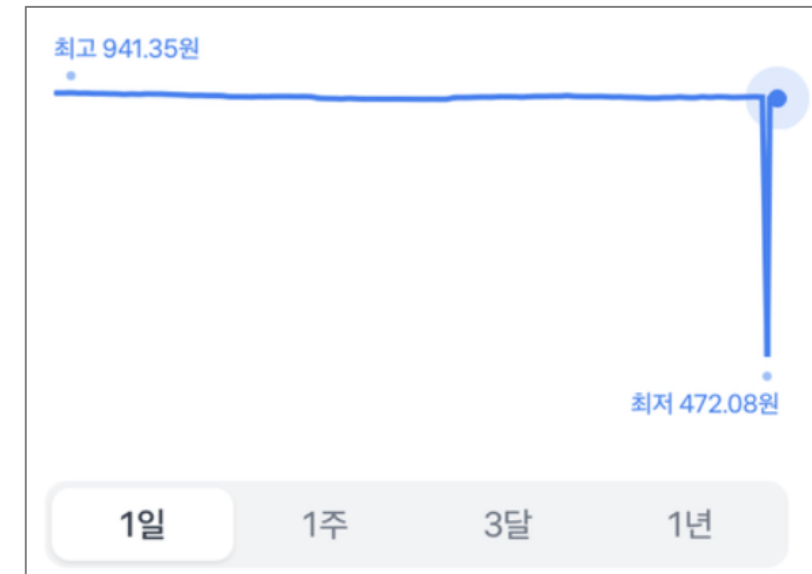
```
<University>
  <Student ID="1">
    <Name>John</Name>
    <Age>18</Age>
    <Degree>B.Sc.</Degree>
  </Student>
  <Student ID="2">
    <Name>David</Name>
    <Age>31</Age>
    <Degree>Ph.D. </Degree>
  </Student>
  ....
</University>
```

Diversity of Data

□ Data types

- Tabular data
 - Table-like structures with specified rows (*instances*) and columns (*attributes*)
 - e.g., relational database table, excel files
- Sequence data
 - The order or sequence of elements
 - e.g., text, biological sequences, shopping transaction sequences
- Time-series data
 - Ordered set of numerical values at specific time intervals
 - e.g., stock market prices, energy consumption, sensor readings
- Spatial, temporal, and spatiotemporal data
 - Spatial (location-based) and temporal (time-based) information
 - e.g., GPS trajectories, traffic data, weather data
- Graph and network data
 - Complex relationships and connections between entities
 - e.g., social networks, communication networks, biological networks

Rank	Name	Market Cap	Price	Today
1	 NVIDIA NVDA	\$4.451 T	\$183.14	▼ 1.54%
2	 Apple AAPL	\$3.759 T	\$255.76	▼ 1.94%
3	 Alphabet (Google) GOOG	\$3.667 T	\$303.21	▼ 1.69%
4	 Microsoft MSFT	\$2.986 T	\$401.86	▼ 0.75%
5	 Amazon AMZN	\$2.249 T	\$209.53	▼ 1.47%



Diversity of Data

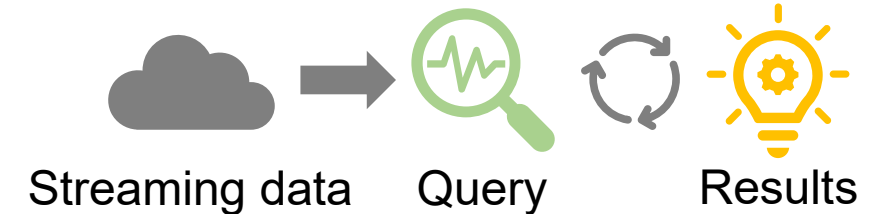
□ Data states

■ Stored data

- a.k.a. offline data, static data
- Finite, stored in various kinds of large data repositories
- Given data, apply queries (tasks) to get a result
- e.g., Databases

■ Streaming data

- a.k.a. online data, dynamic data
- Infinite, constantly coming, real-time response
- Given a query (a task), apply data to get results
- e.g., Social media feeds, IoT sensor data, Video surveillance



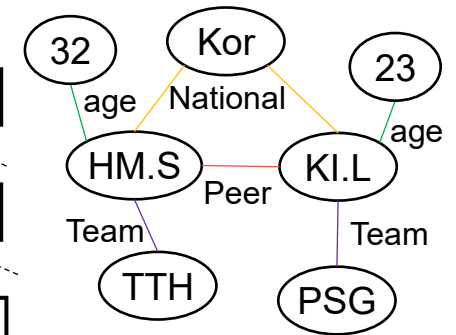
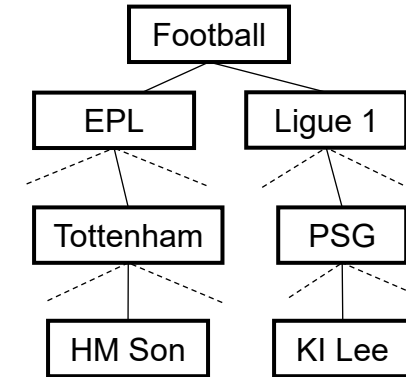
Diversity of Data

Data structure (flexibility-efficiency trade-off)

Heung-Min Son, born on July 8, 1992 (age 32), is a South Korean professional footballer who plays as a forward for Tottenham Hotspur in the Premier League and serves as the ...

```
<Player>
<Name>Heung-Min Son
</Name> <DateOfBirth>1992-07-08 </DateOfBirth>
<Age>32</Age><Nationality>South Korean</Nationality>
<Club> <Team>Tottenham Hotspur</Team> <League>Premier League</League>
</Club></Player>
```

ID	1	2
Name	Heung-Min Son	Kangin Lee
Date of Birth	July 8, 1992	Feb 19, 2001
Age	32	23
Nationality	South Korean	South Korean
Club Team	Tottenham Hotspur	PSG
League	Premier League	Ligue 1



Heung-Min Son with rumors of a transfer to Barcelona, Real Madrid, Man United, ...

Heung-Min Son nets 4th goal of season in loss with Chelsea ...

86' Heung-Min Son wins a free kick in the defensive zone

Data types

Outline

❑ Diversity of data

❑ Understanding data

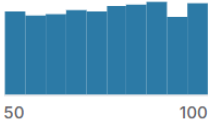
- Data sets, objects, attributes, matrix
- Data statistics
- Data visualization
- Data similarity

❑ Preprocessing data



Understanding Data > Data Sets

□ A data set represents a collection of data (i.e., a city)

△ Student_ID	△ First_Name	△ Last_Name	✉ Email	△ Gender	# Age	△ Department	# Attendance (%)	# Midterm_Score	# Final_Score
Unique identifier for each student	Student's first name	Student's last name	Contact email	Male, Female	The age of the student	Student's department	Attendance percentage 0-100	Midterm exam score (out of 100)	Final exam score
5000 unique values	Maria	Johnson	5000 unique values	Male		CS			
	Ahmed	Jones		Female		Engineering			
	Other (3692)	Other (3282)				Other (1509)			
S1000	Omar	Williams	student0@university.com	Female	22	Engineering	52.29	55.03	57.82
S1001	Maria	Brown	student1@university.com	Male	18	Engineering	97.27	97.23	45.8
S1002	Ahmed	Jones	student2@university.com	Male	24	Business	57.19	67.05	93.68
S1003	Omar	Williams	student3@university.com	Female	24	Mathematics	95.15	47.79	80.63
S1004	John	Smith	student4@university.com	Female	23	CS	54.18	46.59	78.89
S1005	Liam	Brown	student5@university.com	Male	21	Engineering		78.85	43.53
S1006	Ahmed	Jones	student6@university.com	Male	24	Business	57.6	66.26	89.07
S1007	Ahmed	Smith	student7@university.com	Male	19	Engineering	51.91	45.67	73.96
S1008	Omar	Smith	student8@university.com	Female	21	CS	85.97	84.42	90.87

Understanding Data > Data Objects

- A data object represents an entity in a data set (i.e., a citizen in a city)
 - a.k.a., data point, sample, example, instance, tuple
 - Examples
 - Sales database: customers, store items, sales
 - Medical database: patients, hospitals, treatments
 - University database: students, professors, courses
 - Data objects are described by attributes
 - Typically, rows → objects; columns → attributes

ID	Name	Age	...	Salary
1	Kim	24	...	100
2	Jason	35	...	300
...	...	...	...	.
100	Lee	22	...	200

Understanding Data > Data Attributes

- An attribute represents a characteristic of a data object (i.e., info of a citizen)
 - a.k.a., dimension, feature, variable
 - Example: gender, name, grade, age of a customer

A list of COVID-19 test results

ID	Gender	COVID	Name	Grade	Body Temp	Age	...
1	Male	Positive	Kim	Junior	38.5	24	...
2	Female	Negative	Sarah	Graduated	36.0	35	...
...			...	...	.	...	...
100	Female	Negative	Park	Sophomore	36.5	22	...

Understanding Data > Data Attributes

□ An attribute represents a characteristic of a data object

▪ Types

- Categorical (qualitative)
 - Binary (e.g., {true, false})
 - > Symmetric: both states are equally important (e.g., gender: {male, female})
 - > Asymmetric: only one state is important (e.g., COVID: {positive, negative})
 - Nominal (e.g., name)
 - Ordinal (e.g., grade: {freshman, sophomore, junior, senior})

ID	Gender	COVID	Name	Grade	Body Temp	Age	...
1	Male	Positive	Kim	Junior	38.5	24	...
2	Female	Negative	Sarah	Graduated	36.0	35	...
...			...	...	.	...	...
100	Female	Negative	Park	Sophomore	36.5	22	...

Understanding Data > Data Attributes

□ An attribute represents a characteristic of a data object

▪ Types

• Numerical (quantitative)

– Interval-scaled

> No inherent zero-point, values are measured on a scale of equal-sized units (e.g., temperature in °C or °F, IQ scores)

– Ratio-scaled

> Inherent true zero-point (e.g., temperature in Kelvin, length, counts, monetary quantities)

ID	Gender	COVID	Name	Grade	Body Temp	Age	...
1	Male	Positive	Kim	Junior	38.5	24	...
2	Female	Negative	Sarah	Graduated	36.0	35	...
...			...	...	.	...	...
100	Female	Negative	Park	Sophomore	36.5	22	...

Understanding Data > Data Attributes

□ An attribute represents a characteristic of a data object

▪ Ranges

- Discrete: a finite or countably infinite set of values
 - e.g., zip codes, profession, a bag of words
- Continuous: real numbers
 - e.g., temperature, height, or weight

ID	Gender	COVID	Name	Grade	Body Temp	Age	...
1	Male	Positive	Kim	Junior	38.5	24	...
2	Female	Negative	Sarah	Graduated	36.0	35	...
...			...	...	.	...	...
100	Female	Negative	Park	Sophomore	36.5	22	...

Understanding Data > Data Matrix

□ Data matrix (a set of data objects)

- A data matrix of n data points with l dimensions (i.e., attributes)

ID	Gender	COVID	Name	Grade	Body Temp	Age	...
1	Male	Positive	Kim	Junior	38.5	24	...
2	Female	Negative	Sarah	Graduated	36.0	35	...
...			...	...	.	...	...
100	Female	Negative	Park	Sophomore	36.5	22	...

$$\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1l} \\ x_{21} & x_{22} & \dots & x_{2l} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{nl} \end{pmatrix}$$

■ Understanding Data > Statistics

❑ Central tendency

- Mean, median, mode

❑ Dispersion

- Range, quartile, five-number summary, outlier

❑ Covariance and correlation

Understanding Data > Statistics

□ Central tendency

▪ Mean: an average value of data

- Arithmetic mean
- Weighted mean
- Trimmed mean
 - Chopping extreme values (e.g., dropping max/min scores)

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\bar{x} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i}$$

$$\bar{x} = \frac{\sum_{j=2}^{n-1} w_j x_j}{\sum_{j=2}^{n-1} w_j}$$

(x_j : ordered values)

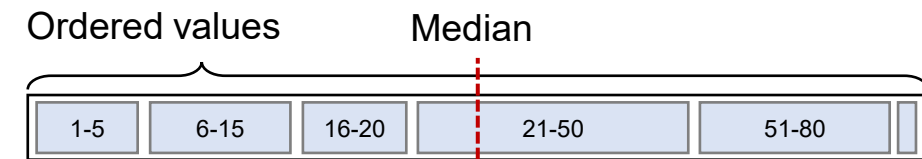
▪ Median: a middle value of ordered data

- Odd number of values → middle value
- Even number of values → average of the middle two values
- Grouped data → approximate median by interpolation

the lower boundary of the median interval The total number of values Sum of frequencies lower than the median interval

$$median \approx L_1 + \left(\frac{N/2 - (\sum freq)_l}{freq_{median}} \right) width$$

Frequency of the median interval Width of the median interval



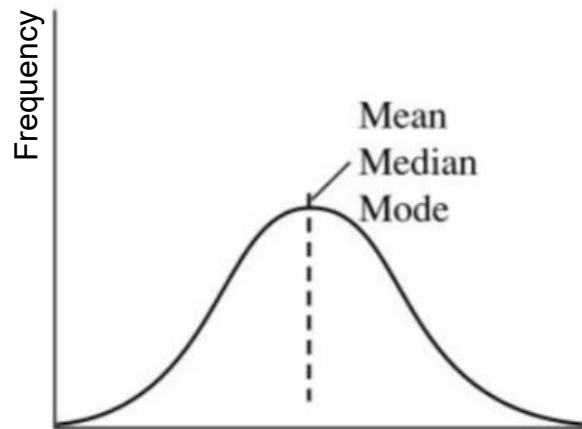
<i>age</i>	<i>frequency</i>
1-5	200
6-15	450
16-20	300
21-50	1500
51-80	700
81-110	44

$$21 + \left(\frac{\frac{3194}{2} - 950}{1500} \right) 30 = 33.94$$

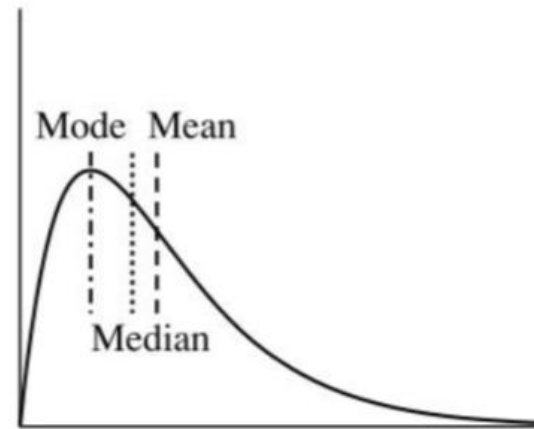
Understanding Data > Statistics

□ Central tendency

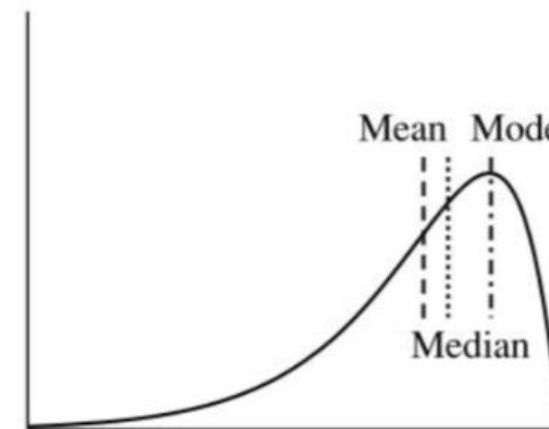
- Mode: a value that occurs most frequently
 - Unimodal
 - For moderately skewed data, empirically, $mean - mode \approx 3 \times (mean - median)$
 - Positively skewed: $Mode < Median < Mean$
 - Negatively skewed: $Mean < Median < Mode$
 - Multi-modal
 - Greatest frequencies found in several different values. e.g., bimodal and trimodal



(a) Symmetric data



(b) Positively skewed data

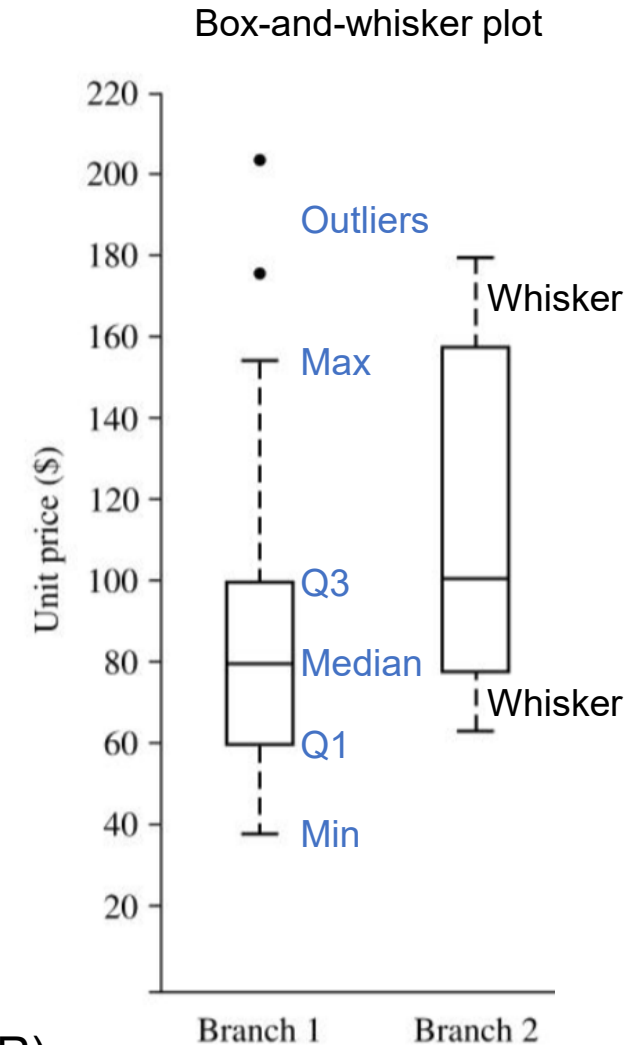
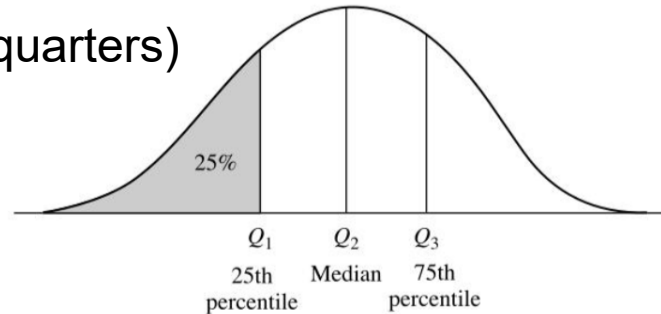


(c) Negatively skewed data

Understanding Data > Statistics

□ Dispersion

- Range
 - Max value – min value
- Quartile
 - Divide an ordered data into four parts (quarters)
 - Q1(25th percentile)
 - Q2(50th percentile = median)
 - Q3(75th percentile)
 - Interquartile range
 - $IQR = Q3 - Q1$
- Five number summary
 - Min, Q1, Median, Q3, Max
- Outlier
 - A value lower/higher than 1.5 IQR (e.g., $x < Q1 - 1.5 \cdot IQR$ or $x > Q3 + 1.5 \cdot IQR$)



Understanding Data > Statistics

□ Dispersion

▪ Variance and standard deviation

- A measure of how far a set of numbers is spread out from their average value
- Variance: the expected value of the squared deviation from the mean of values
- Standard deviation: the square root of the variance

$$\mu = E(X) = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\sigma^2 = \text{var}(X) = E[(X - \mu)^2] = \begin{cases} \sum_x (x - \mu)^2 f(x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx & \text{if } X \text{ is continuous} \end{cases}$$

$$\sigma^2 = E[(X - \mu)^2] = E[X^2] - \mu^2 = E[X^2] - [E(X)]^2$$

Understanding Data > Statistics

□ Covariance and correlation

▪ Covariance

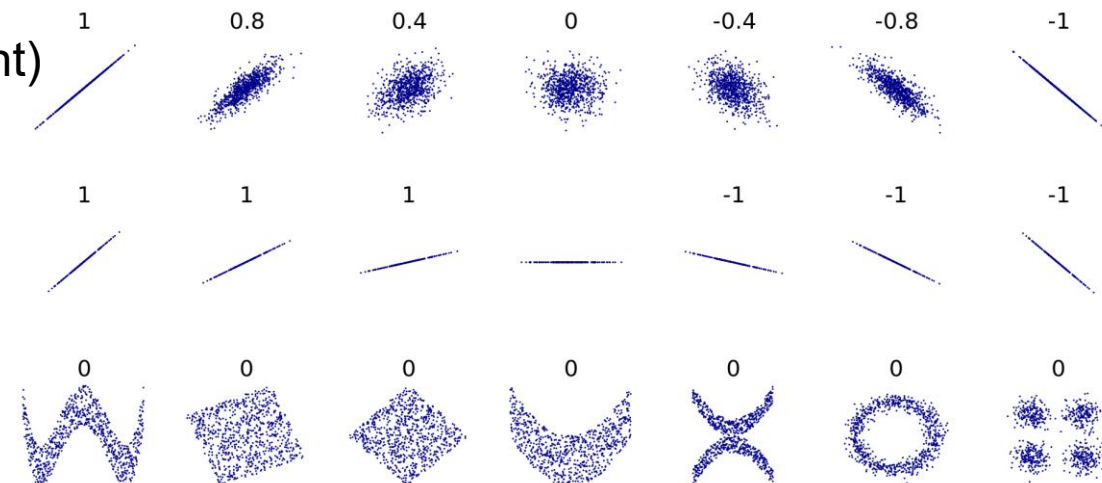
- A measure of how much two variables (X and Y) change together

$$\sigma_{XY} = E[(X - \mu_X)(Y - \mu_Y)] = E[XY] - \mu_X\mu_Y$$

▪ Correlation coefficient

- a.k.a. Pearson's correlation coefficient
- $\rho_{XY} > 0$: positive correlation
- $\rho_{XY} < 0$: negative correlation
- $\rho_{XY} = 0$: no linear correlation (e.g., may be independent)

$$\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y} = \frac{E[(X - \mu_X)(Y - \mu_Y)]}{\sqrt{E[X^2] - \mu_X^2} \sqrt{E[Y^2] - \mu_Y^2}}$$



■ Understanding Data > Visualization

□ Distribution plot

- Histogram
- Density plot
- Box plot
- Violin plot

□ Scatter plot

□ Quantile plot

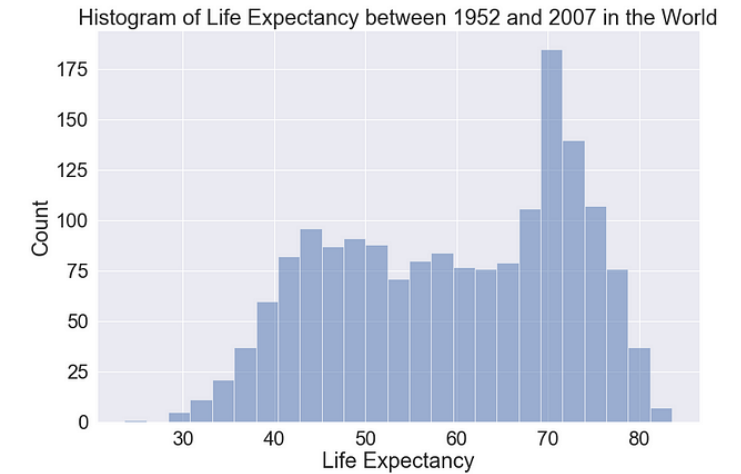
- Quantile-Quantile plot

Understanding Data > Visualization

□ Distribution plot

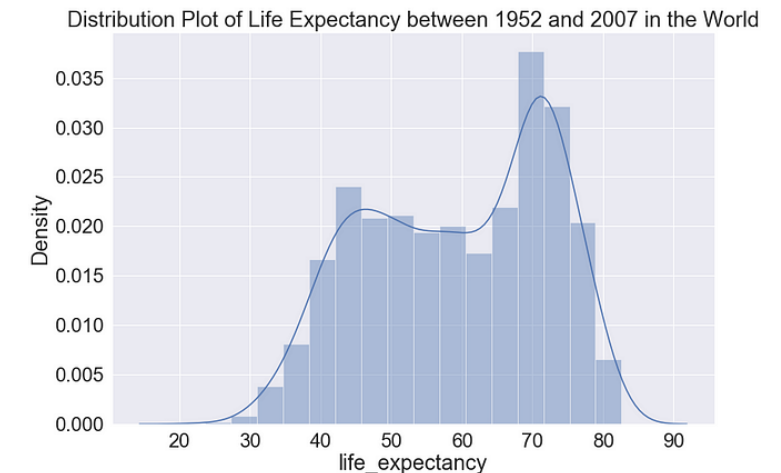
■ Histogram

- Shows the overall (discrete) distribution of values
- Divides the range of values into intervals (i.e., buckets or bins)
 - i.e., ranges in X-axis
- Displays the frequency of observations in each bin (i.e., bars)
 - i.e., values in Y-axis



■ Density plot

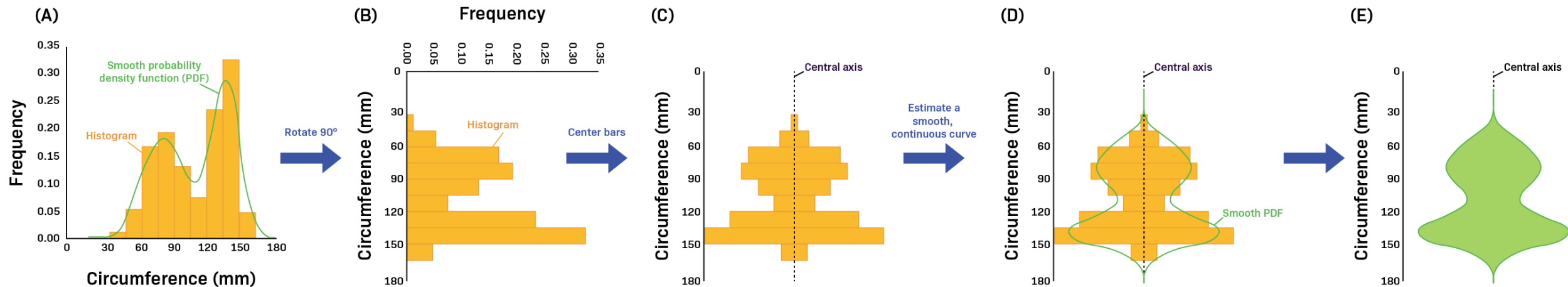
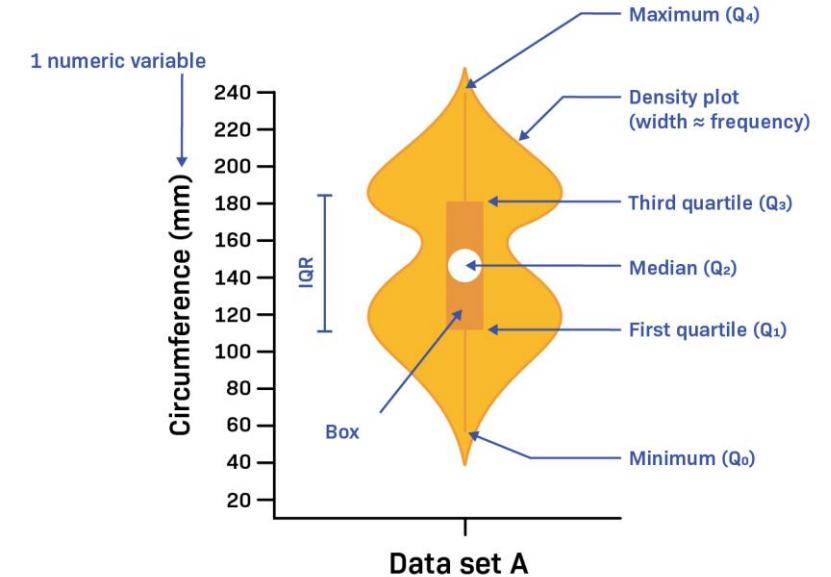
- Shows the overall (continuous) distribution of values
- Smoothing curve is used instead of bars
- A continuous estimate of the probability density function
 - i.e., Kernel density estimation



Understanding Data > Visualization

□ Distribution plot

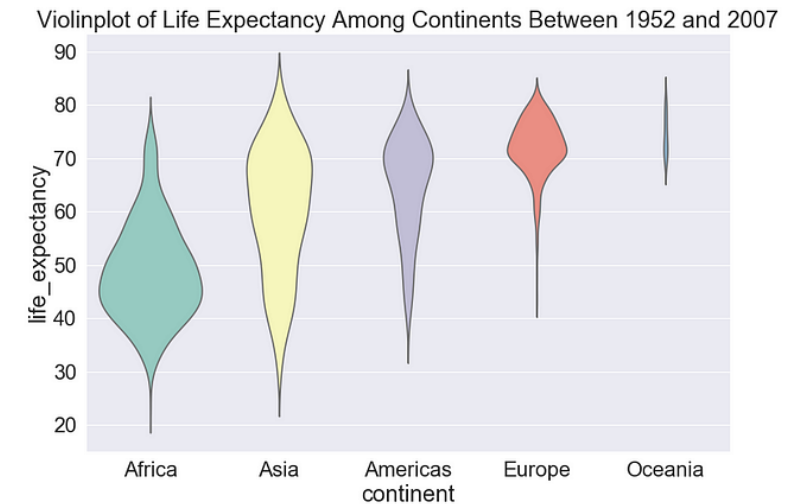
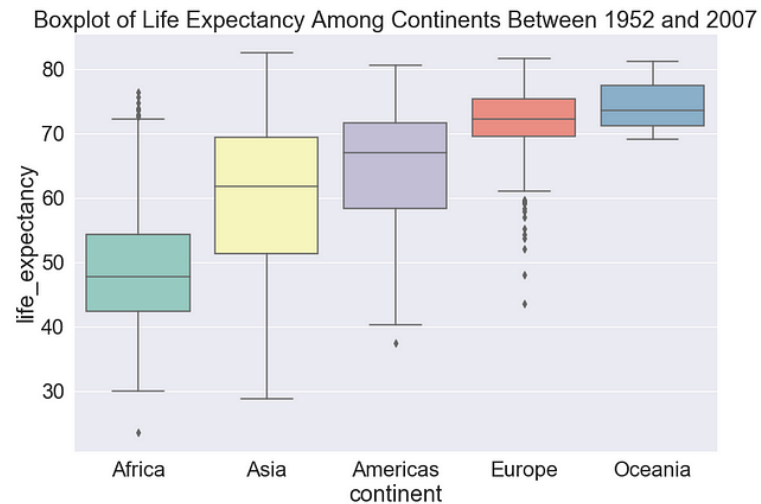
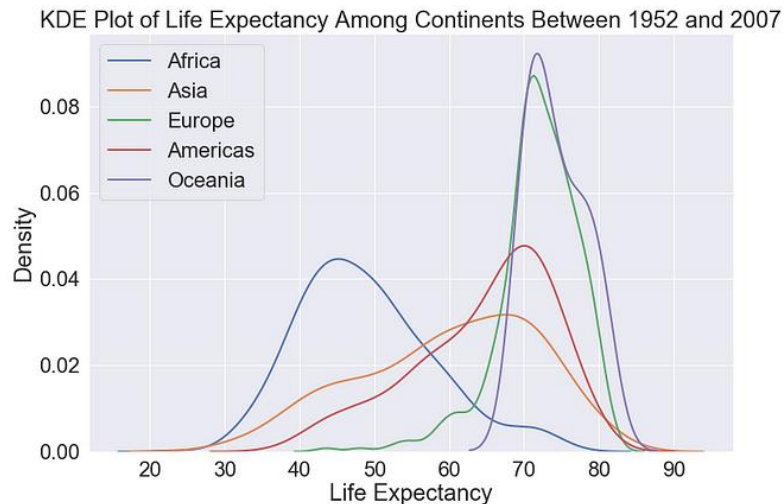
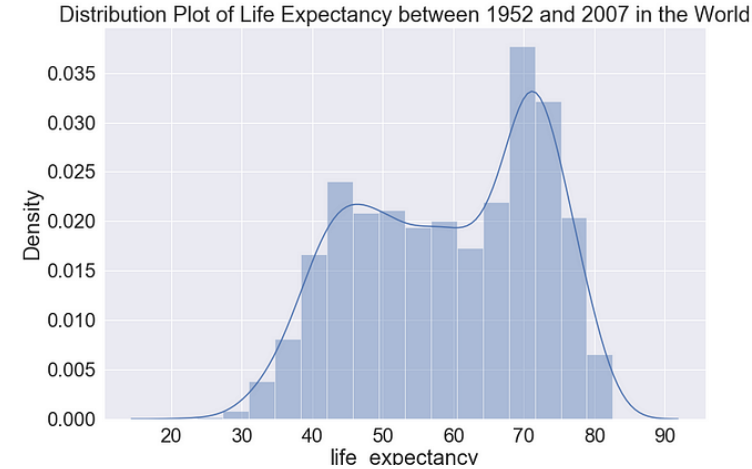
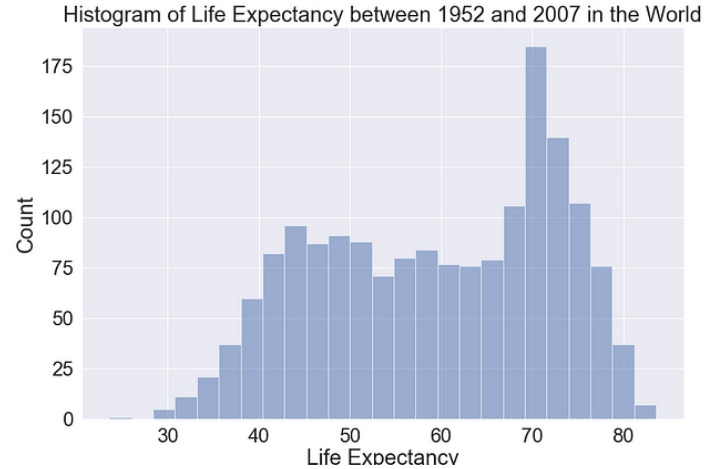
- Box plot (a.k.a. box-and-whisker plot)
 - Min, Q1, Median, Q3, Max
- Violin plot
 - Combines box plot and density plot
 - Allows for richer comparison of different distributions
 - Procedures



Understanding Data > Visualization

□ Distribution plot

■ Example

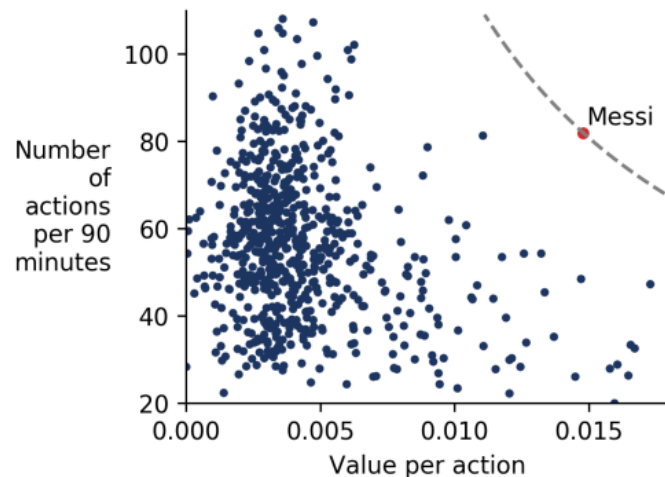


Understanding Data > Visualization

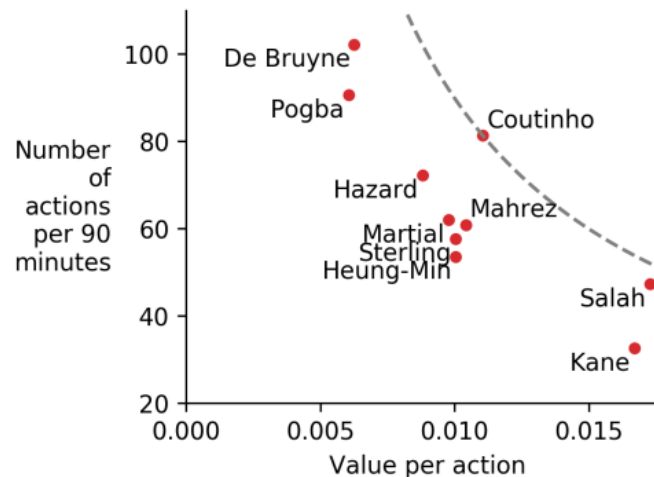
□ Scatter plot

- Determines whether there appears to be a relationship, pattern, or trend in or between data sets
- Visualizes relationships between (usually) two continuous variables
 - Each pair of values is treated as a pair of coordinates and plotted as points in the plane
- Shows correlation between the two variables and/or comparison between observations

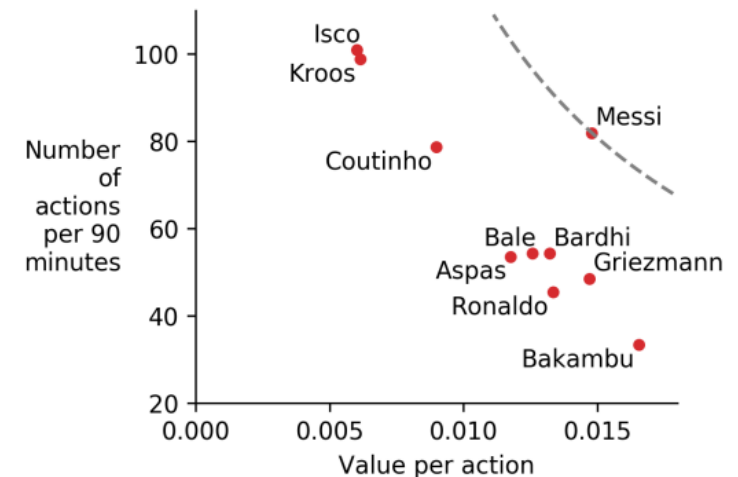
Players in the 2017/2018 season who played at least 900 minutes in the Spanish or English league



(a) All players



(b) Top-10 players in the English league



(c) Top-10 players in the Spanish league

Understanding Data > Visualization

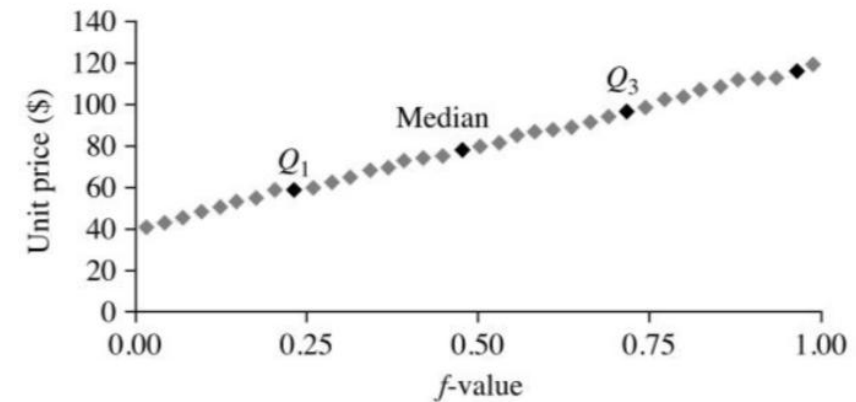
Quantile plot

- Shows a quantile (equal-sized subsets) information of a single data set
 - 4-quantile → quartile, 100-quantile → percentile
- Each ordered observation x_i is associated with a quantile f_i
 - Reorganize the observations into ordered set with an equal-width
 - e.g., $f_i = \frac{i-0.5}{N}$ (N is the number of data)

Unit price (\$)
75
120
40
...
78
115



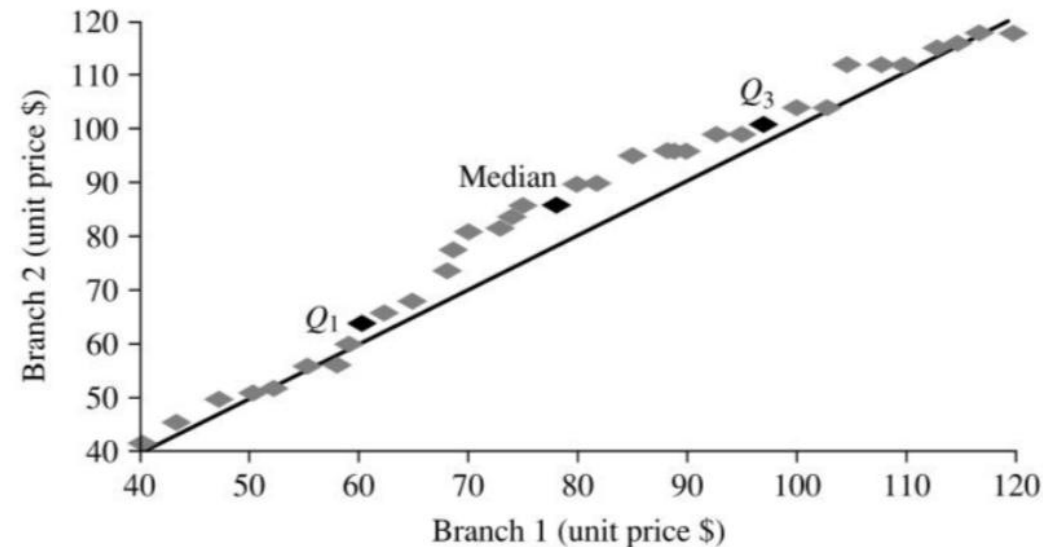
i	f_i	Unit price (\$)
1	0.013	40
2	0.040	43
3	0.067	47
...	...	...
36	0.959	117
37	0.986	120



Understanding Data > Visualization

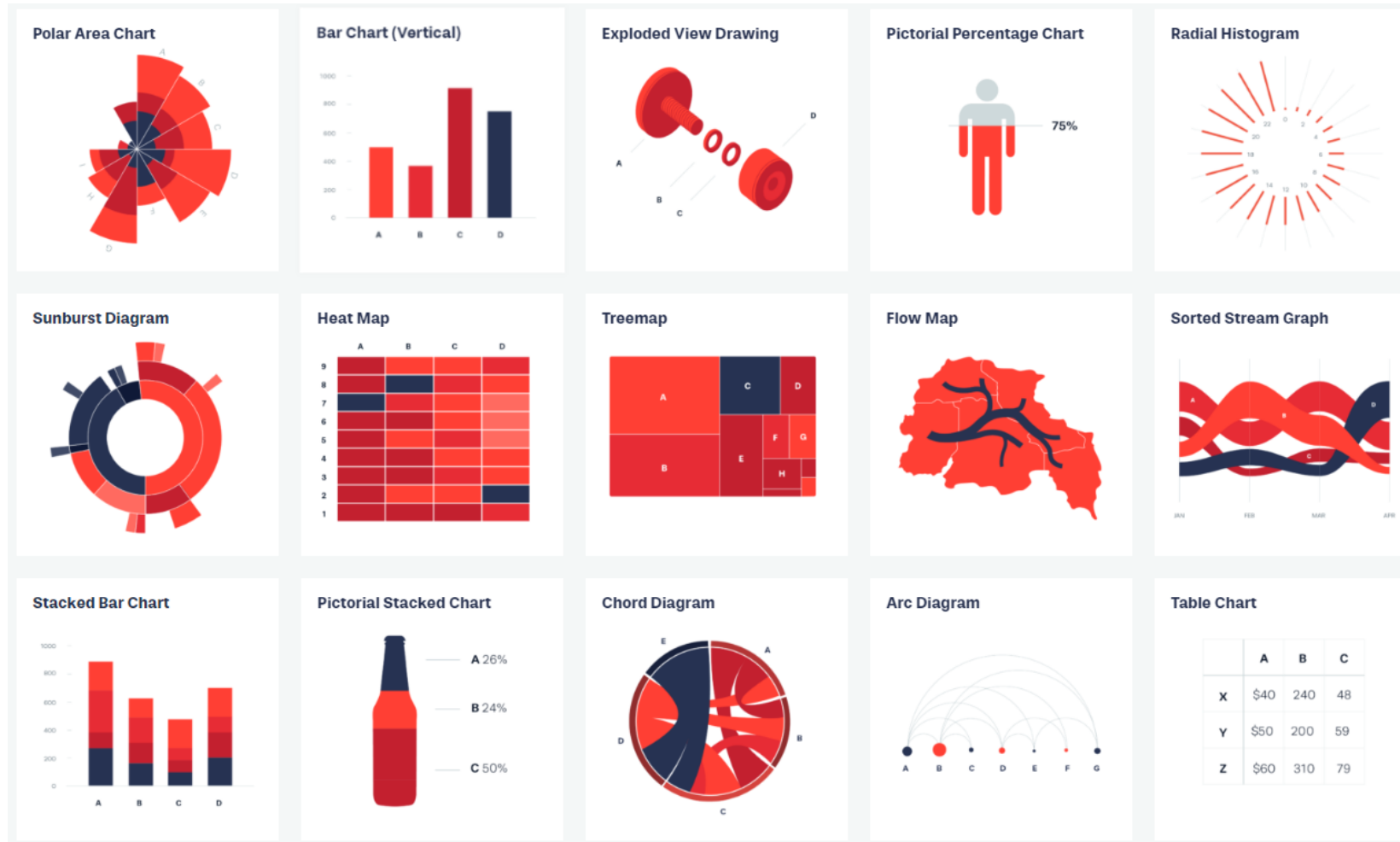
Quantile-Quantile plot

- Easily compare two data sets by comparing their quantiles
 - i.e., match observations in the two data sets by their rankings
- Example
 - The unit price of items sold in Branch 1 tends to be lower than those in Branch 2
 - Such trends are more highlighted in the Interquartile range (IQR)



Understanding Data > Visualization

□ There are numerous options for more graphical visualization!



Understanding Data > Visualization

❑ 1 dataset 100 visualizations?

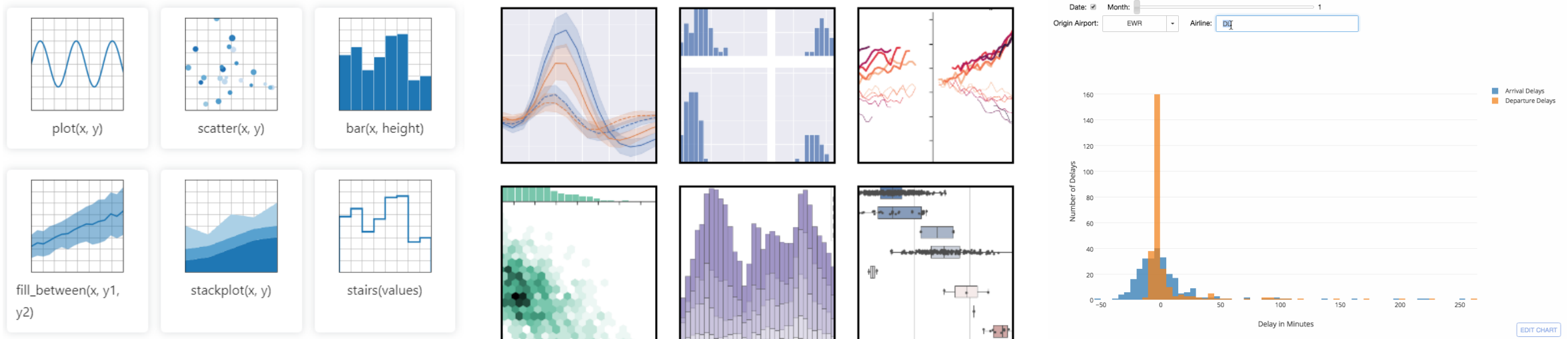
Number of World Heritage Sites			
	Norway	Denmark	Sweden
2004	5	4	13
2022	8	10	15

<https://100.datavizproject.com/>

Understanding Data > Visualization

□ Popular Python libraries

- Matplotlib (<https://matplotlib.org/>)
 - Most popular in the scientific community. Customization of various plot types
- Seaborn (<https://seaborn.pydata.org/>)
 - Built on top of Matplotlib and provides a high-level interface
- Plotly (<https://plotly.com/python/>)
 - Interactive web-based visualization



■ Understanding Data > Similarity

□ Similarity

- A real-valued function that quantifies the similarity between two objects
- Measure how two data objects are alike: the higher value, the more alike
- Often falls in the range $[0,1]$: 0: no similarity; 1: completely similar

□ Dissimilarity (i.e., distance)

- Numerical measure of how different two data objects are
- In some sense, the inverse of similarity: the lower, the more alike
- Minimum dissimilarity is often 0 (i.e., completely similar)
- Range $[0, 1]$ or $[0, \infty)$, depending on the definition

□ Proximity usually refers to either similarity or dissimilarity

Understanding Data > Similarity

□ (Recap) Data matrix

- A data matrix of n data points with l dimensions (i.e., attributes)

$$\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1l} \\ x_{21} & x_{22} & \dots & x_{2l} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{nl} \end{pmatrix}$$

□ Similarity/distance matrix

- Similarity or distance of two data points x_i and x_j by a function $d(\cdot, \cdot)$
- Usually symmetric, thus a triangular matrix

$$\begin{pmatrix} d(x_1, x_1) & & & \\ d(x_2, x_1) & d(x_2, x_2) & & \\ \vdots & \vdots & \ddots & \\ d(x_n, x_1) & d(x_n, x_2) & \dots & d(x_n, x_n) \end{pmatrix}$$

Understanding Data > Similarity > Numerical Attributes

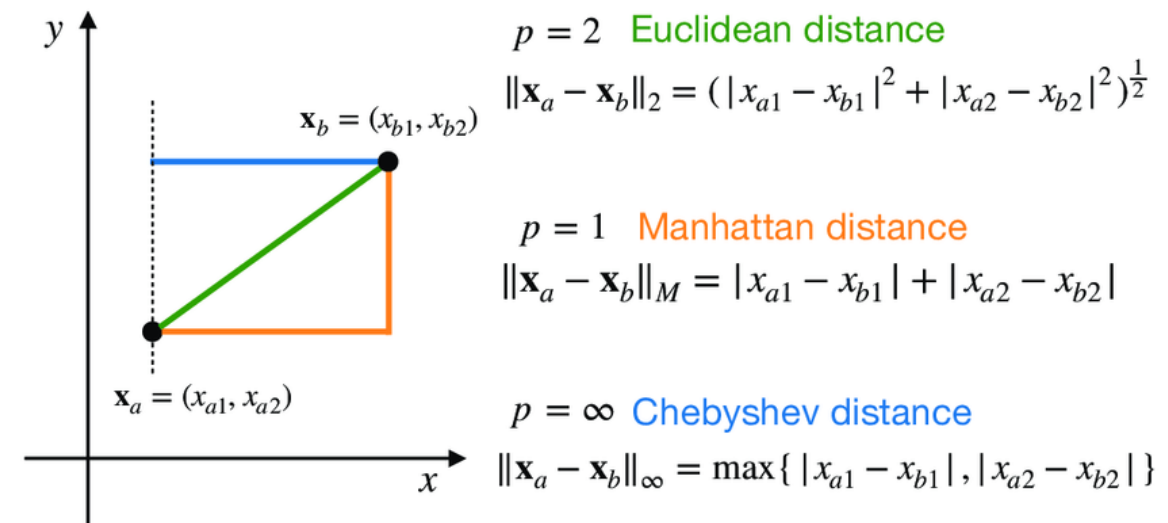
□ Properties for a distance function (i.e., metric)

- Positivity: $d(i, j) > 0$ if $i \neq j$, and $d(i, i) = 0$
- Symmetry: $d(i, j) = d(j, i)$
- Triangle inequality: $d(i, j) \leq d(i, k) + d(k, j)$

□ Minkowski distance

- A generalized popular distance metric
- Special cases
 - $p = 1$ (L1 norm): Manhattan (or city block) distance
 - e.g., Hamming distance
 - $p = 2$ (L2 norm): Euclidean distance
 - $p \rightarrow \infty$ ($L_{\max}$ norm or $L_{\inf}$ norm): Supremum distance
 - e.g., Chebyshev distance

$$d(x_i, x_j) = \sqrt[p]{|x_{i1} - x_{j1}|^p + |x_{i2} - x_{j2}|^p + \cdots + |x_{il} - x_{jl}|^p}$$



Understanding Data > Similarity > Binary Attributes

□ Contingency table for binary attributes

- Count the number of states for each data objects

□ Symmetric binary distance

- Two states are equally important
- e.g., Gender: {Male, Female}

□ Asymmetric binary distance

- One state (usually positive) is more important
- e.g., Covid case: {Positive, Negative}

□ Jaccard coefficient (index or similarity)

- A popular similarity measure for asymmetric binary attributes

		Object i	
Object j		1	0
	1	q	r
	0	s	t

$$d(i, j) = \frac{r + s}{q + r + s + t}$$

$$d(i, j) = \frac{r + s}{q + r + s}$$

$$J(i, j) = 1 - \frac{r + s}{q + r + s} = \frac{q}{q + r + s}$$

Understanding Data > Similarity > Binary Attributes

□ Example

- Asymmetric binary distance (except Gender)?
 - Y, P → 1 (important) and N → 0 (not important)

Name	Gender	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	M	Y	N	P	N	N	N
Jim	M	Y	Y	N	N	N	N
Mary	F	Y	N	P	N	P	N

- Out of the three people, which two are the most dissimilar to each other?
 - Jack and Jim? Jim and Mary? Jack and Mary?

Understanding Data > Similarity > Binary Attributes

□ Example

- Asymmetric binary distance (except Gender)?
 - Y, P → 1 (important) and N → 0 (not important)

Name	Gender	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	M	Y	N	P	N	N	N
Jim	M	Y	Y	N	N	N	N
Mary	F	Y	N	P	N	P	N

		Mary		
Jack		1	0	$\sum_{\text{row}}$
	1	2	0	2
	0	1	3	4
	$\sum_{\text{col}}$	3	3	6

$$d(\text{jack}, \text{mary}) = \frac{0 + 1}{2 + 0 + 1} = 0.33$$

Understanding Data > Similarity > Binary Attributes

□ Example

- Asymmetric binary distance (except Gender)?
 - Y, P → 1 (important) and N → 0 (not important)

Name	Gender	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	M	Y	N	P	N	N	N
Jim	M	Y	Y	N	N	N	N
Mary	F	Y	N	P	N	P	N

		Mary		
Jack		1	0	$\sum_{\text{row}}$
	1	2	0	2
	0	1	3	4
	$\sum_{\text{col}}$	3	3	6

Jack

	1	0	Σ_{row}
1	1	1	2
0	1	3	4
Σ_{col}	2	4	6

$$d(\text{jack}, \text{mary}) = \frac{0 + 1}{2 + 0 + 1} = 0.33$$

$$d(\text{jack}, \text{jim}) = \frac{1 + 1}{1 + 1 + 1} = 0.67$$

Understanding Data > Similarity > Binary Attributes

□ Example

- Asymmetric binary distance (except Gender)?
 - Y, P $\rightarrow$ 1 (important) and N $\rightarrow$ 0 (not important)

Name	Gender	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	M	Y	N	P	N	N	N
Jim	M	Y	Y	N	N	N	N
Mary	F	Y	N	P	N	P	N

		Mary		
Jack		1	0	$\sum_{\text{row}}$
	1	2	0	2
	0	1	3	4
	$\sum_{\text{col}}$	3	3	6

Jim

	1	0	$\sum_{\text{row}}$
1	1	1	2
0	1	3	4
$\sum_{\text{col}}$	2	4	6

Jack

		Mary		
		1	0	Σ_{row}
Jim	1			
	0			
	Σ_{col}			

$$d(\text{jack}, \text{mary}) = \frac{0 + 1}{2 + 0 + 1} = 0.33$$

$$d(\text{jack}, \text{jim}) = \frac{1 + 1}{1 + 1 + 1} = 0.67$$

$$d(\text{jim}, \text{mary}) =$$

Understanding Data > Similarity > Binary Attributes

□ Example

- Asymmetric binary distance (except Gender)?
 - Y, P $\rightarrow$ 1 (important) and N $\rightarrow$ 0 (not important)

Name	Gender	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	M	Y	N	P	N	N	N
Jim	M	Y	Y	N	N	N	N
Mary	F	Y	N	P	N	P	N

		Mary		
Jack		1	0	$\sum_{\text{row}}$
	1	2	0	2
	0	1	3	4
	$\sum_{\text{col}}$	3	3	6

$$d(\text{jack}, \text{mary}) = \frac{0 + 1}{2 + 0 + 1} = 0.33$$

		Jim		
		1	0	$\sum_{\text{row}}$
Jack	1	1	1	2
	0	1	3	4
	$\sum_{\text{col}}$	2	4	6

$$d(\text{jack}, \text{jim}) = \frac{1 + 1}{1 + 1 + 1} = 0.67$$

		Mary		
		1	0	$\sum_{\text{row}}$
Jim	1	1	1	2
	0	2	2	4
	$\sum_{\text{col}}$	3	3	6

$$d(\text{jim}, \text{mary}) = \frac{1 + 2}{1 + 1 + 2} = 0.75$$

Understanding Data > Similarity > Nominal Attributes

❑ Simple matching

- See how many values of attributes are matched
- m: # of matches, p: total # of attributes

$$d(i, j) = \frac{p - m}{p}$$

❑ Convert to a large number of asymmetric binary attributes

- Creating a new binary attribute for each of the states (i.e., one-hot encoding)
- Calculate asymmetric binary distances

Original Data			One-Hot Encoded Data				
	Team	Points		Team_A	Team_B	Team_C	Points
Obj1	A	25		1	0	0	25
Obj2	A	12		1	0	0	12
Obj3	B	15		0	1	0	15
Obj4	B	14		0	1	0	14
Obj5	B	19		0	1	0	19
Obj6	B	23		0	1	0	23
Obj7	C	25		0	0	1	25
Obj8	C	29		0	0	1	29



Asymmetric binary distance

Obj2			
	1	0	$\sum_{\text{row}}$
Obj3	1	0	1
	0	1	2
$\sum_{\text{col}}$	1	2	3

$$d(\text{obj2}, \text{obj3}) = \frac{1 + 1}{0 + 1 + 1} = 1.0$$

Understanding Data > Similarity > Ordinal Attributes

□ Order is important

- e.g., {freshman, sophomore, junior, senior}

□ Can be treated like numeric attributes such as interval-scaled

- Replace an ordinal attribute value given N states by its rank $\rightarrow r_i \in \{1, \dots, N\}$

- Normalize the rank onto $[0, 1] \rightarrow z_i = \frac{r_i - 1}{N - 1}$

- e.g.,

- freshman: $(1-1)/(4-1) = 0$
- sophomore: $(2-1)/(4-1) = 1/3$
- junior: $(3-1)/(4-1) = 2/3$
- senior $(4-1)/(4-1) = 1$

- Compute the distance using any methods for numerical attributes

- e.g., $d(\text{freshman}, \text{senior}) = 1$, $d(\text{junior}, \text{senior}) = 1/3$

Understanding Data > Similarity > Mixed Types

- ❑ A dataset may contain all attribute types
 - Nominal, symmetric binary, asymmetric binary, numeric, and ordinal
- ❑ A weighted distance to combine their effects
 - Distance between objects i and j with p attributes (f) of mixed types

$$d(i, j) = \frac{\sum_{f=1}^p w_{ij}^{(f)} d_{ij}^{(f)}}{\sum_{f=1}^p w_{ij}^{(f)}}$$

Object Identifier	Test-1 (nominal)	Test-2 (ordinal)	Test-3 (numeric)
1	code A	excellent → 1.0	45
2	code B	fair → 0.0	22
3	code C	good → 0.5	64
4	code A	excellent → 1.0	28

$$d_{i=1,j=2}^{(f=\text{Test1})} = 1 \quad d_{i=1,j=2}^{(f=\text{Test2})} = 1.0 \quad d_{i=1,j=2}^{(f=\text{Test3})} = \frac{45-22}{64-22} = 0.55$$

$$\text{Assuming } w_{ij}^{(f=\text{Test1})} = 0.2, w_{ij}^{(f=\text{Test2})} = 0.3, w_{ij}^{(f=\text{Test3})} = 0.5, \quad d(i, j) = \frac{0.2 \times 1 + 0.3 \times 1.0 + 0.5 \times 0.55}{0.2 + 0.3 + 0.5} = 0.775$$

- Notes
 - Weight of an attribute f , $w_{ij}^{(f)} = 0$ if x_{if} and x_{jf} are missing or not important (asymmetric binary attributes)
 - $d_{ij}^{(f)}$ needs to be normalized into $[0, 1]$

Understanding Data > Similarity > Vectors

□ High-dimensional and sparse vector

- Only none-zero elements needs to be considered
- e.g., document term-frequency, user-item interaction

Document	Team	Coach	Hockey	Baseball	Soccer	Penalty	Score	Win	Loss	Season
Document1	5	0	3	0	2	0	0	2	0	0
Document2	3	0	2	0	1	1	0	1	0	1
Document3	0	7	0	2	1	0	0	3	0	0
Document4	0	1	0	0	1	2	2	0	3	0

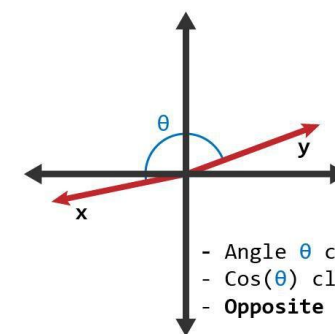
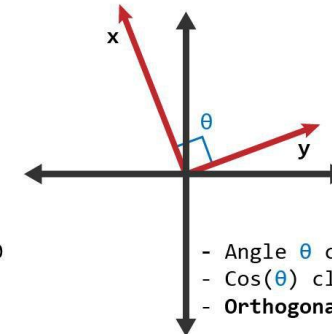
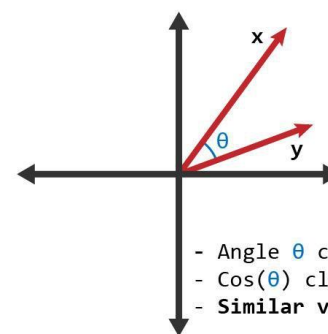
□ Cosine similarity

- Measures the similarity between two vectors in an inner product space
- The cosine of the angle showing whether they are pointing in the same direction

$$\cos(x_1, x_2) = \frac{x_1 \cdot x_2}{\|x_1\| \times \|x_2\|} \sim [-1, 1]$$

$\|x\|$: Euclidean norm (i.e., length) of a vector x

$x_1 \cdot x_2$: Dot product of vectors x_1 and x_2



- Cosine distance = 1 – cosine similarity

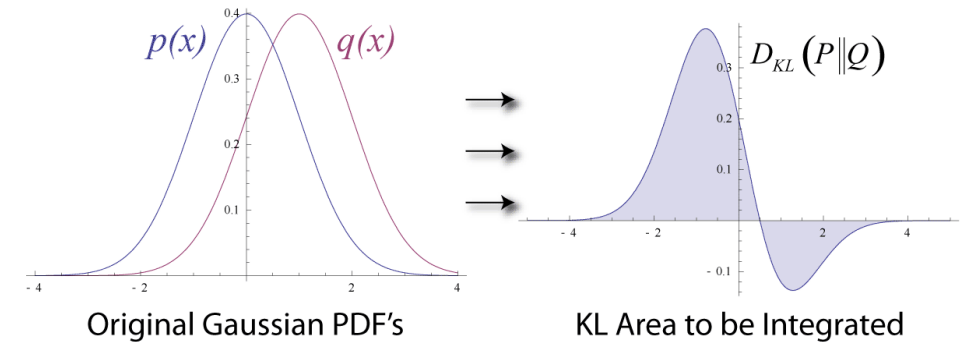
Understanding Data > Similarity > Distributions

□ Kullback-Leibler (KL) divergence

- Measure the difference between two probability distributions over the same variable x
- From information theory, closely related to relative entropy, information divergence, and information for discrimination
- $D_{KL}(p(x) || q(x))$: divergence of $q(x)$ from $p(x)$
 - Measuring the information lost when $q(x)$ is used to approximate $p(x)$

$$D_{KL}(p(x)||q(x)) = \sum_{x \in X} p(x) \ln \frac{p(x)}{q(x)}$$

$$D_{KL}(p(x)||q(x)) = \int_{-\infty}^{\infty} p(x) \ln \frac{p(x)}{q(x)} dx$$

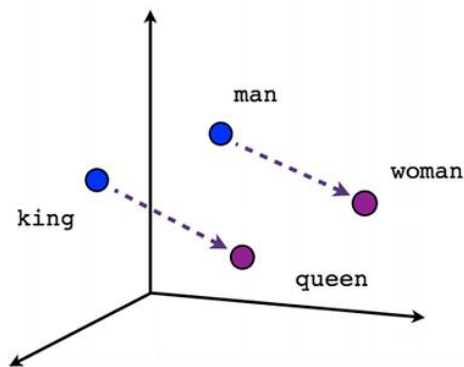


- Not a distance measure: asymmetric, not satisfy triangular inequality
 - $D_{KL}(P||Q)$ does not equal $D_{KL}(Q||P)$

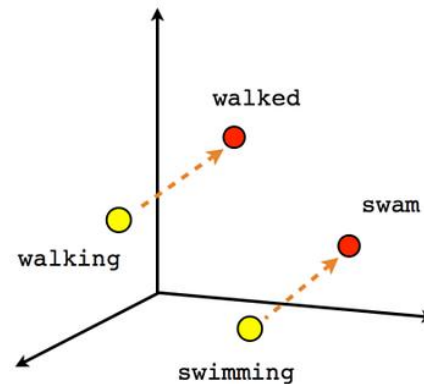
Understanding Data > Similarity

□ Capturing hidden semantics in distance

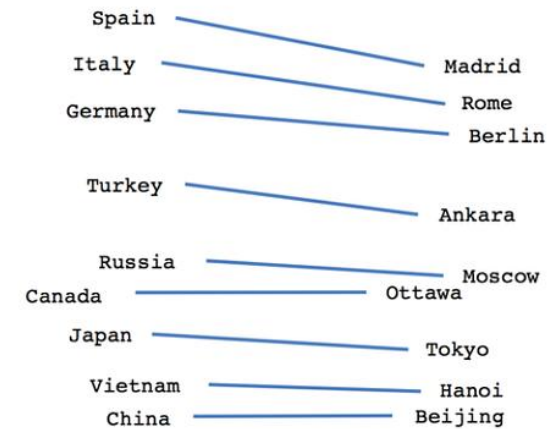
- The above similarity measures cannot capture hidden semantics
 - e.g., which pairs are more similar: geometry, algebra, music, politics?
- The same bags of words may express rather different meanings
 - e.g., “The cat bites a mouse” vs. “The mouse bites a cat”
- New similarity measures may be needed to handle complex semantics
 - e.g., Distributive representation and representation learning



Male-Female



Verb tense



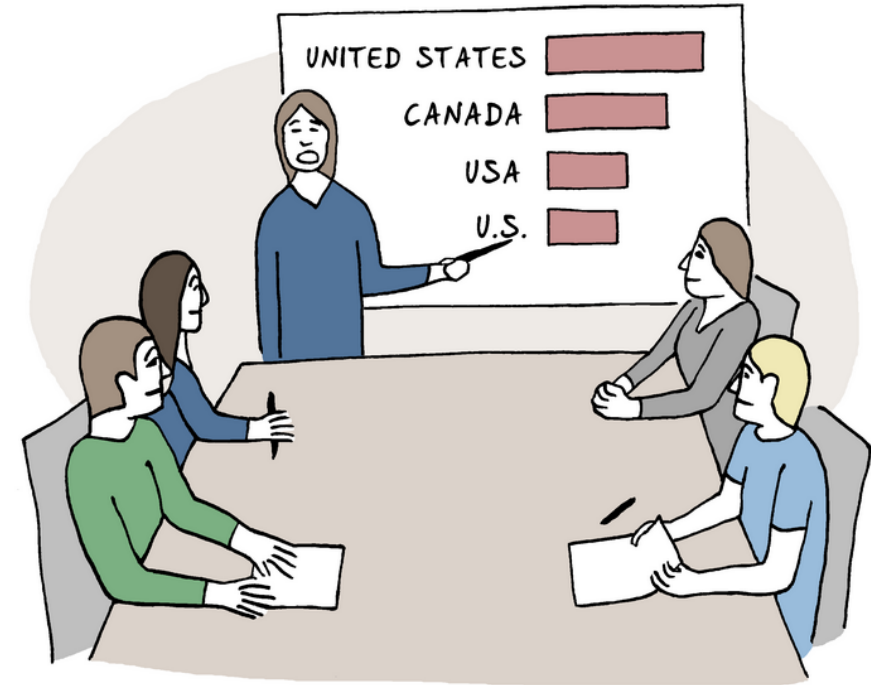
Country-Capital

Semantle

Reference

Outline

- ❑ Diversity of data
- ❑ Understanding data
- ❑ **Preprocessing data**
 - Data quality measures
 - Data cleaning
 - Data integration
 - Data transformation



AS YOU CAN SEE, OUR TOP MARKETS ARE
UNITED STATES, CANADA, USA AND THE U.S.

■ Preprocessing Data > Quality Measures

☐ Accuracy

- Is the data value correct or incorrect?

☐ Completeness

- Have all the data been recorded? or available?

☐ Consistency

- Has some of data been modified while some not?

☐ Timeliness

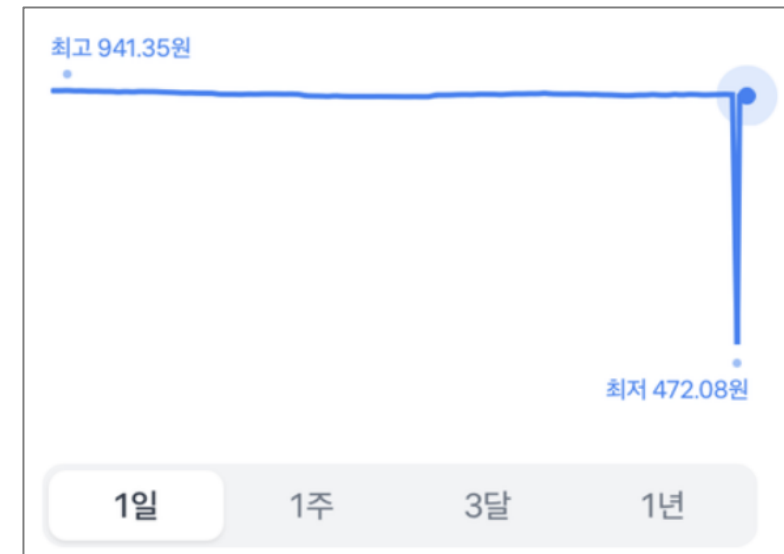
- Has the data been timely updated?

☐ Believability

- How trustable the data are correct?

☐ Interpretability

- How easily the data can be understood?



■ Preprocessing Data > Cleaning

- ❑ Cleaning dirty data with lots of potentially incorrect data
 - Possible reasons: instrument faulty, human or computer error, and transmission error
- ❑ Issues
 - Missing (or incomplete) values
 - Lacking attribute values of interest or containing only aggregate data
 - e.g., Sales = “ ” or Sales = “100” (average value)
 - Noisy values
 - Containing noise, errors, or outliers
 - e.g., Salary = “-10”
 - Inconsistent values
 - Containing discrepancies in records
 - e.g., Rating = “B” or “2” ({A, B, C} or {1, 2, 3})
 - Intentional values (i.e., disguised missing values)
 - e.g., Birthday = “Jan. 1” (as everyone’s birthday)

■ Preprocessing Data > Cleaning

- ❑ Cleaning dirty data with lots of potentially incorrect data
 - Possible reasons: instrument faulty, human or computer error, and transmission error
- ❑ Example
 - Are there any issues with this data set?

Customer ID	Age	Email	Subscription	Sales
1	10	kim@naver.com	Active	100
2	-5	jane#google.com	Expired	
3	32	NA	Pending	210
4	45	park@korea.ac.kr	active	270
new	50	dave@korea.ac.kr	Cancelled	120
...	...	...	...	...

Preprocessing Data > Cleaning

❑ Cleaning dirty data with lots of potentially incorrect data

- Possible reasons: instrument faulty, human or computer error, and transmission error

❑ Example

- Are there any issues with this data set?

The diagram illustrates data cleaning issues with arrows pointing to specific cells in the table:

- Noisy**: Points to the Age value -5 for Customer ID 2.
- Inconsistent**: Points to the Email value jane#google.com for Customer ID 2.
- Intentional**: Points to the Subscription value active for Customer ID 4.
- Missing**: Points to the Sales value 120 for Customer ID 5.

Customer ID	Age	Email	Subscription	Sales
1	10	kim@naver.com	Active	100
2	-5	jane#google.com	Expired	
3	32	NA	Pending	210
4	45	park@korea.ac.kr	active	270
5	new	dave@korea.ac.kr	Cancelled	120
...	...	...	...	...

■ Preprocessing Data > Cleaning > Missing

❑ MCAR (Missing Completely At Random)

- The probability of a value being missing is essentially a coin flip, unrelated to any other variable.
- e.g., a sensor randomly loses power for 5 minutes due to an external power grid error.

❑ MAR (Missing At Random)

- Missingness depends on other observed variables in the dataset, not the missing value itself.
- e.g., men might be less likely to fill out the “skincare routine” survey question than women.

(If we know the gender, we might predict the missingness.)

❑ MNAR (Missing Not At Random)

- Missingness depends heavily on the unobserved missing value itself.
- e.g., people with high debt are less likely to report their credit card balance on a loan application.

■ Preprocessing Data > Cleaning > Missing

□ Handling missing values

- Fill in the missing value manually
 - Optimal, but tedious and infeasible
- Ignore the object with missing values (e.g., for MCAR)
 - e.g., remove objects with missing class labels when doing classification
 - Not effective when the ratio of missing values per attribute varies considerably
- Fill in it automatically with (e.g., for MAR)
 - The statistics (e.g., mean) of existing attribute values (in the same class if available)
 - Inference the most probable values
 - Regression, nearest neighbors, decision tree, or Bayesian methods
 - e.g., the annual salary of a person can be inferred using his occupation and age
- Indicate missingness with a flag (e.g., for MNAR)
 - Fill in it with a global constant: e.g., “unknown” as a new class when doing classification

ID	Age	Email	Subscription	Sales
3	32	NA	Pending	210

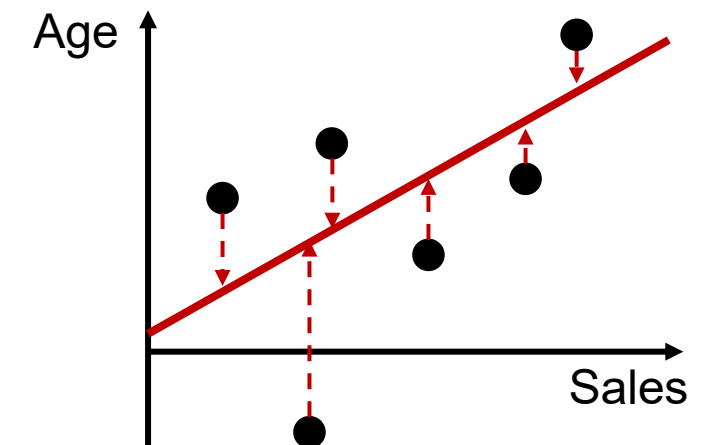
Sales
100
210
270
120
...

Preprocessing Data > Cleaning > Noisy

□ Handling noisy values

- Binning
 - Sort values and partition into equal-frequency or equal-width bins
 - Smooth original values by bin means, median, or bin boundaries, etc.
- Regression
 - Smooth by fitting the data into regression functions
- Clustering
 - Detect and remove outliers
- Filtering
 - Consult predefined rules (e.g., age should be positive integer)

	<i>Equal-width bin</i>	<i>Mean</i>
Age	Age	Age
10	0-10	2.5
-5	0-10	2.5
32	31-40	32
45	41-50	47.5
50	41-50	47.5



■ Preprocessing Data > Cleaning > Inconsistent/Intentional

□ Handling inconsistent/intentional values

- Simply correct it with a predefined rule
 - e.g., new = max(customer ID) + 1
- Filter by regular expression (i.e., regex)
 - e.g., Email address: `^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$`
- Inspect the unique values
 - e.g., Subscription = [Active, active, Expired, Pending, Cancelled]

Customer ID	Age	Email	Subscription	Sales
1	10	kim@naver.com	Active	100
2	-5	jane#google.com	Expired	
3	32	NA	Pending	210
4	45	park@korea.ac.kr	active	270
new	50	dave@korea.ac.kr	Cancelled	120
...	...	...	...	...

Preprocessing Data > Integration

❑ Combining data from multiple sources into a coherent data store

▪ Advantages

- Help reduce or avoid noise
- Get a more complete picture of data
- Improve speed and quality of analysis

▪ Issues

- Redundancy
- Inconsistency

▪ Methods

- Schema integration
 - Integrate metadata from different sources (e.g., $A.C\# \equiv B.ID$)
- Entity resolution
 - Identify the matching records from multiple sources (e.g., Gil Dong Hong $\equiv$ Hong Gil Dong)
- Correlation analysis
 - Keep only one of two attributes showing a high correlation (e.g., $A.Deposit = 2 * B.Sales$)

Dataset A

C#	Name	Deposit
1	Kim	200
2	Jane	240
4	Park	540
5	Dave	240
...	...	



Dataset B

ID	Age	Email	Subscription	Sales
1	2.5	kim@naver.com	Active	100
2	2.5	jane@google.com	Expired	120
4	47.5	park@korea.ac.kr	Active	270
5	47.5	dave@korea.ac.kr	Cancelled	120
...	...	...	...	...

■ Preprocessing Data > Transformation

□ Mapping the entire values of an attribute to a new set of replacement values

■ Advantages

- Make the attributes in different units or scales be comparable
- Help improve model performance
- Simplify complex relationships between attributes for easy interpretation

■ Methods

- Normalization
 - Scaled to fall within a smaller, specified range
- Discretization
 - Partition values into coarse intervals (i.e., binning or clustering)
 - Concept hierarchy generation
- Reduction
 - Obtain a reduced representation of a data set

■ Preprocessing Data > Transformation

□ Normalization

- Min-max normalization: to $[new_minA, new_maxA]$

$$v' = \frac{v - minA}{maxA - minA} (new_maxA - new_minA) + new_minA$$

- Example

- Let income values range from \$12,000 to \$98,000 and we want to normalize them to $[0.0, 1.0]$
- Then \$73,600 is mapped to $\frac{73,600 - 12,000}{98,000 - 12,000} (1.0 - 0) + 0 = 0.716$

- Z-score normalization (i.e., standardization) with μ : mean and σ : standard deviation

- The distance between the raw score and the population mean in the unit of the standard deviation
 - The new mean = 0, the new standard deviation = 1

$$v' = \frac{v - \mu}{\sigma}$$

- Example

- Let $\mu = 54000$ and $\sigma = 16000$
- Then, z-score of \$73,600 = $\frac{73,600 - 54,000}{16,000} = 1.225$

Preprocessing Data > Transformation

□ Normalization

- Min-max normalization
- Z-score normalization (standardization)

$$x_{std}^{(i)} = \frac{x^{(i)} - \mu_x}{\sigma_x}$$

standardization

$$x_{norm}^{(i)} = \frac{x^{(i)} - \mathbf{x}_{min}}{\mathbf{x}_{max} - \mathbf{x}_{min}}$$

*min-max scaling
("normalization")*

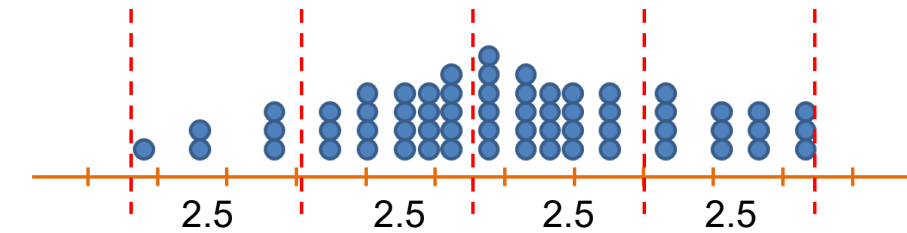
	input	standardized	normalized
0	0	-1.46385	0.0
1	1	-0.87831	0.2
2	2	-0.29277	0.4
3	3	0.29277	0.6
4	4	0.87831	0.8
5	5	1.46385	1.0

Preprocessing Data > Transformation

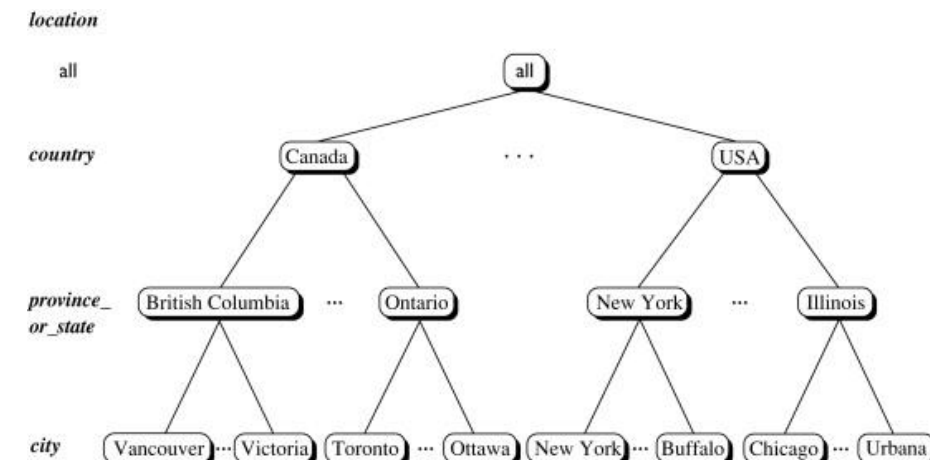
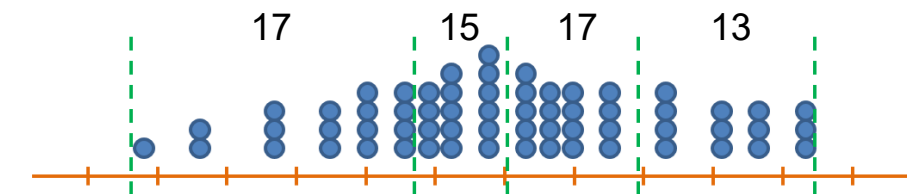
Discretization

- Binning and clustering
 - Transform values into bins or clusters (like handling noisy data)
 - Equal-width: the width of each bin is uniform
 - Equal-frequency: the frequency of each bin is (roughly) constant
- Concept hierarchy generation
 - Specification of a partial/total ordering of attributes explicitly at the schema level by users or experts
 - e.g., street < city < state < country
 - Automatic generation of hierarchies by the analysis of the number of distinct values
 - e.g., {street, city, state, country} → street < city < state < country

Equal Width Discretization



Equal Frequency Discretization



■ Preprocessing Data > Transformation

□ Reduction

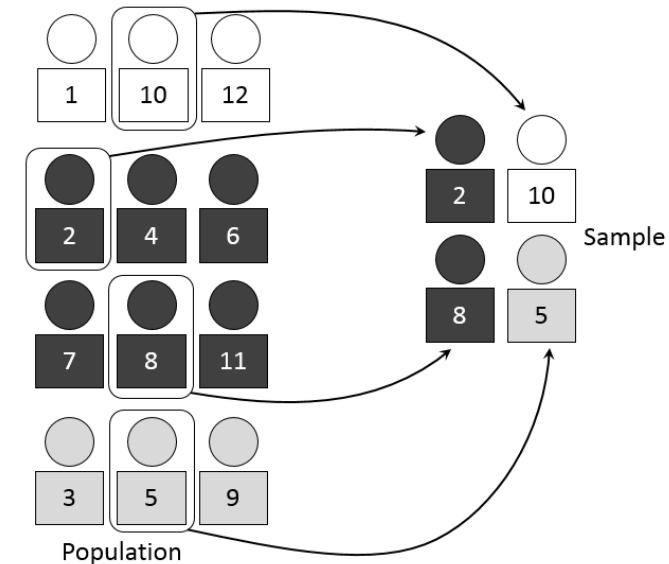
- Sampling
- Compression
- Dimensionality reduction
 - Feature subset selection
 - Linear methods
 - Non-linear methods

■ Preprocessing Data > Transformation > Sampling

□ Reduction

■ Sampling

- Obtaining a set of small samples to represent the whole data set
- Sampling with or without replacement
 - With replacement: consistent and independent samples
 - Without replacement: representative and irredundant samples
- Random sampling
 - Set the equal probability of selecting items
 - May have very poor performance in the presence of skewness
- Stratified sampling
 - Partition the entire data set and draw samples from each partition proportionally
 - Approximately keep the ratio of each partition

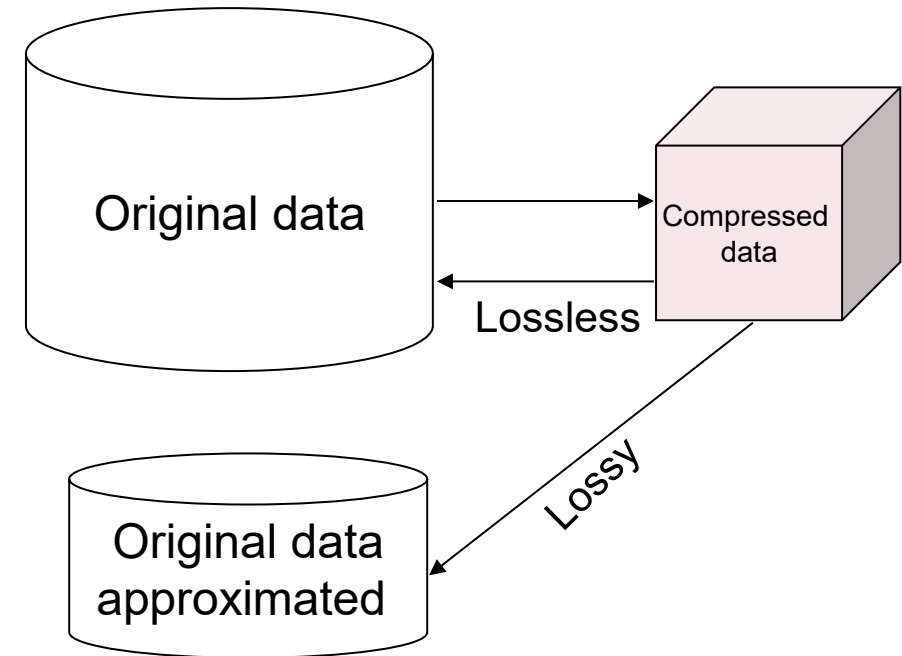


■ Preprocessing Data > Transformation > Compression

□ Reduction

■ Compression

- String compression
 - Typically lossless, but only limited manipulation is possible
 - e.g., Run-Length Encoding (RLE),
 - AAAABBBBCCCCDDDDDD → 4A3B3C5D
- Image/audio/video compression
 - Typically lossy compression, with progressive refinement
 - e.g., PNG, GIF, JPEG, MP3, MP4...
- Time-series compression
 - e.g., Fourier transform, Wavelet transform, ...



Preprocessing Data > Transformation > Dim Reduction

□ Reduction

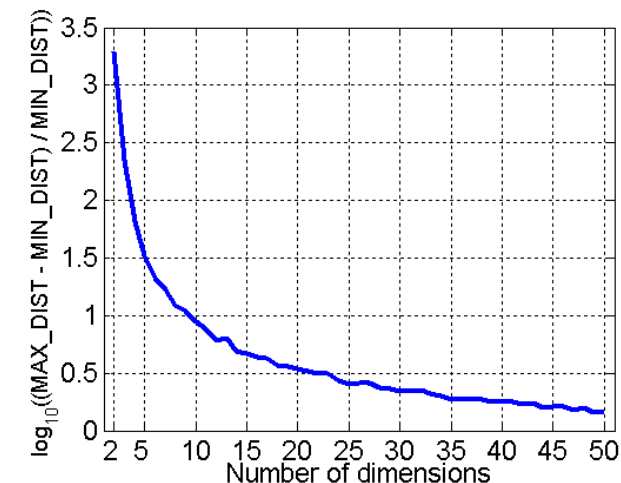
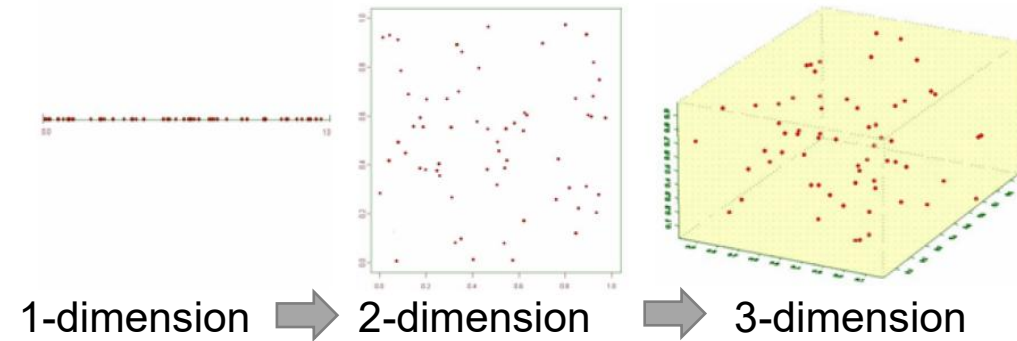
▪ Dimensionality reduction

• Curse-of-dimensionality

- When dimensionality increases, data becomes increasingly sparse in the space that it occupies
- Density and distance between points become meaningless

• Approaches

- Attribute subset selection
- Linear methods
 - > Principal component analysis (PCA)
- Non-linear methods
 - > Kernel principal component analysis (Kernel PCA)
 - > t-Stochastic neighborhood embedding (t-SNE)
 - > Uniform manifold approximation and projection (UMAP)



- Randomly generate 500 points
- Compute difference between max and min distance between any pair of points

■ Preprocessing Data > Transformation > Dim Reduction

□ Attribute subset selection

- Reduce the dimensionality of data by removing redundant and irrelevant features
- Redundant features
 - Duplicating much or all of the information contained in one or more other attributes
 - e.g., the purchase price of a product and the amount of sales tax paid
- Irrelevant features
 - Containing no information that is useful for the data science task at hand
 - e.g., students' ID is often irrelevant to the task of predicting students' GPA
- Heuristics (greedy approach)
 - Hypothesis testing- or value-based
 - Stepwise forward selection: repeatedly picking the best single-attribute
 - Stepwise backward elimination: repeatedly eliminating the worst attribute
 - Combination of the two heuristics
 - Decision tree-based (e.g., information gain-based): using the attributes that appeared in the decision tree

■ Preprocessing Data > Transformation > Dim Reduction

□ Attribute subset selection

▪ Example

Initial attribute set: {A1, A2, A3, A4, A5, A6} Associated value: {20, 3, 7, 40, 10, 20}

Forward selection

Initial reduced set:
{ } / Value: 0

⇒ {A4} / Value: 40

⇒ {A1, A4} / Value: 60

⇒ Reduced attribute set:
{A1, A4, A6} / Value: 80

Backward selection

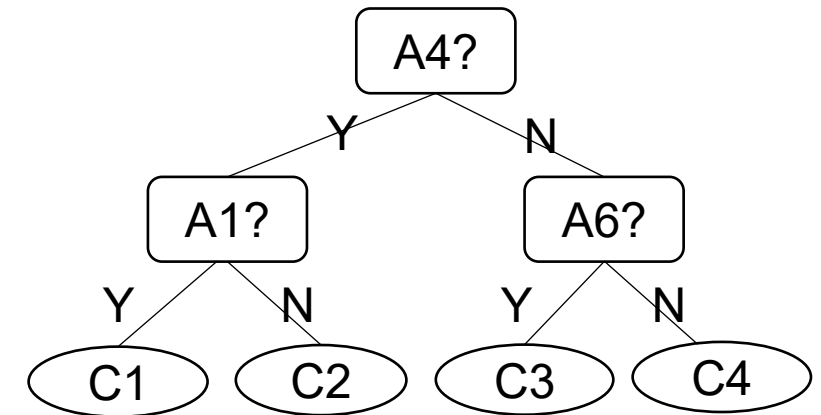
Initial reduced set:
{A1, A2, A3, A4, A5, A6} / Value: 100

⇒ {A1, A3, A4, A5, A6} / Value: 97

⇒ {A1, A4, A5, A6} / Value: 90

⇒ Reduced attribute set:
{A1, A4, A6} / Value: 80

Decision tree induction



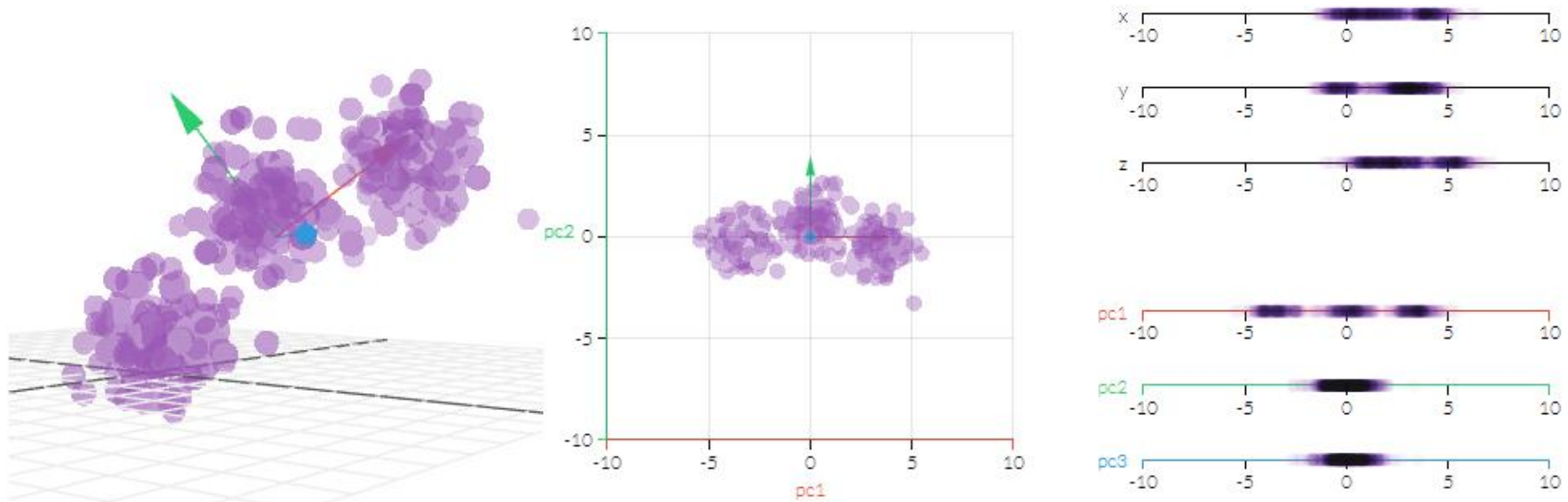
⇒ Reduced attribute set:
{A1, A4, A6}

Preprocessing Data > Transformation > Dim Reduction

□ Linear methods

■ Principal component analysis (PCA)

- A statistical procedure that uses an orthogonal transformation to convert a set of observations of possibly correlated variables into a set of values of linearly uncorrelated variables called principal components
- The number of principal components is less than or equal to the number of original variables

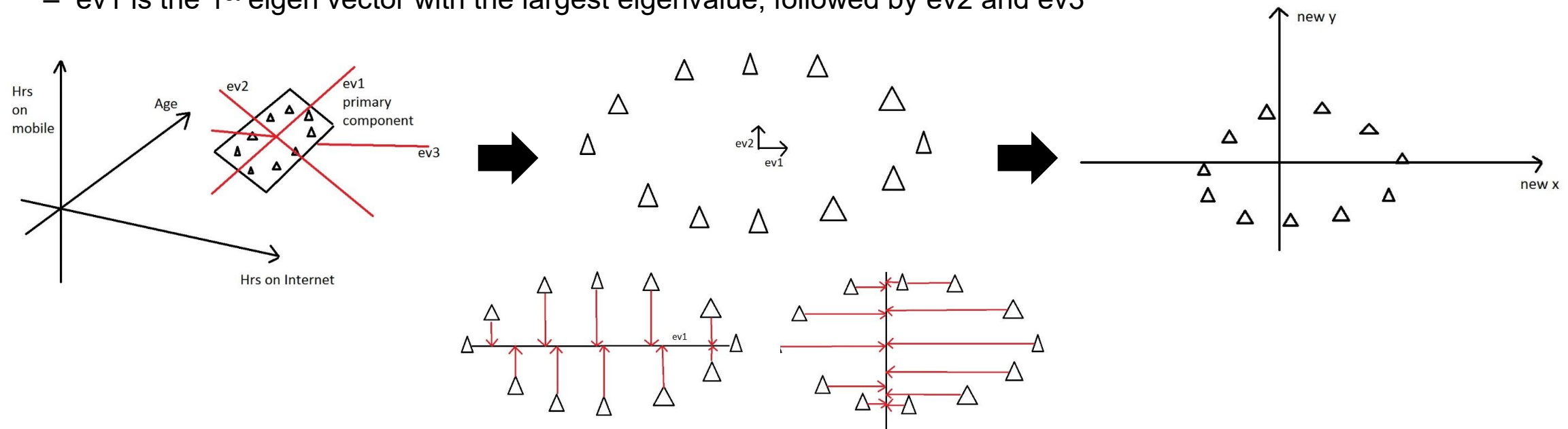


Preprocessing Data > Transformation > Dim Reduction

□ Linear methods

▪ Principal component analysis (PCA)

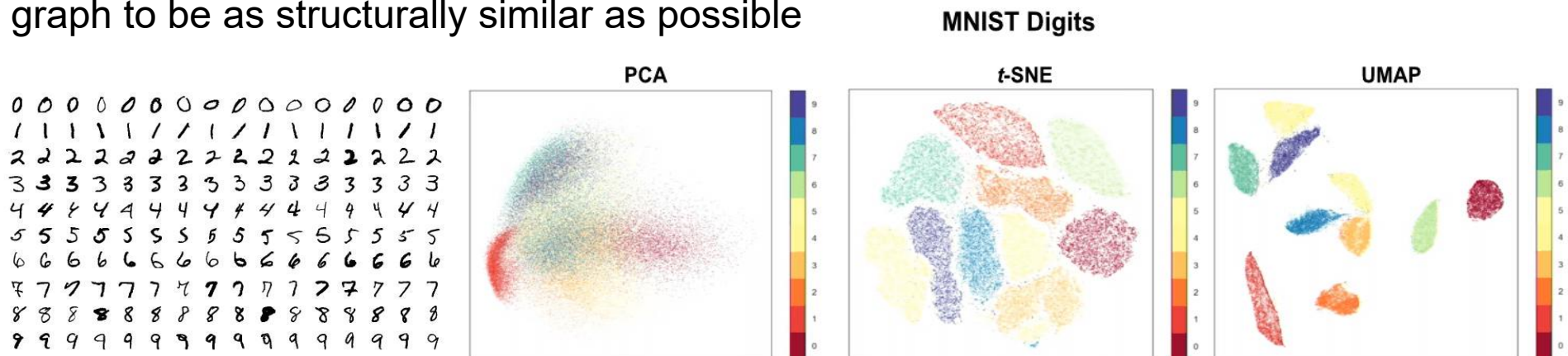
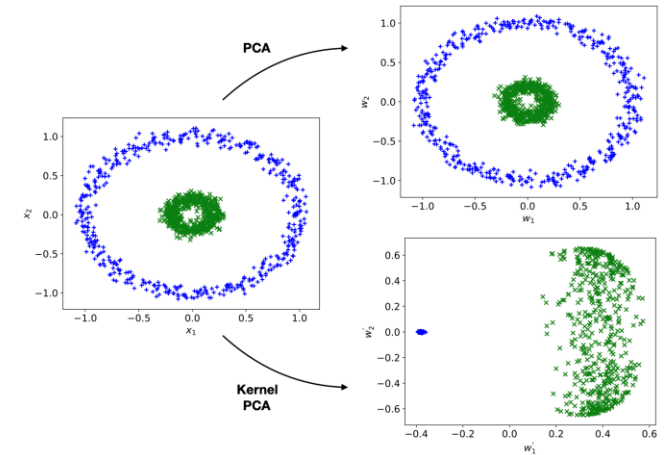
- An eigenvector: a direction of the principal component
- An eigenvalue: a value, telling the variance of the data in that direction
- Example
 - ev1 is the 1st eigen vector with the largest eigenvalue, followed by ev2 and ev3



Preprocessing Data > Transformation > Dim Reduction

□ Non-linear methods <https://projector.tensorflow.org/>

- Kernel PCA
 - Uses kernel functions to project data into a higher-dim space
- t-SNE (t-Distributed Stochastic Neighboring Embedding)
 - Converts similarities between data points to joint probabilities and tries to minimize the divergence between these probabilities in a lower-dim space
- UMAP (Uniform Manifold Approximation and Projection)
 - Constructs a high-dim graph representation of the data and optimizes a lower-dim graph to be as structurally similar as possible



Tutorial References

❑ Basic Python, Jupyter Notebook or Google Colab

- <https://wesmckinney.com/book/python-basics>
- <https://docs.jupyter.org/en/latest/install/notebook-classic.html>
- <https://colab.research.google.com/>

❑ Pandas DataFrame

- <https://wesmckinney.com/book/pandas-basics>
- https://learningds.org/ch/06/pandas_intro.html
 - Datasets: <https://github.com/DS-100/textbook/tree/master/content/datasets>

❑ Preprocessing data

- <https://wesmckinney.com/book/data-cleaning>
- https://learningds.org/ch/09/wrangling_intro.html
 - Datasets: <https://github.com/DS-100/textbook/tree/master/content/datasets>

Outline

□ Today

- Diversity of data
- Understanding Data
- Preprocessing Data

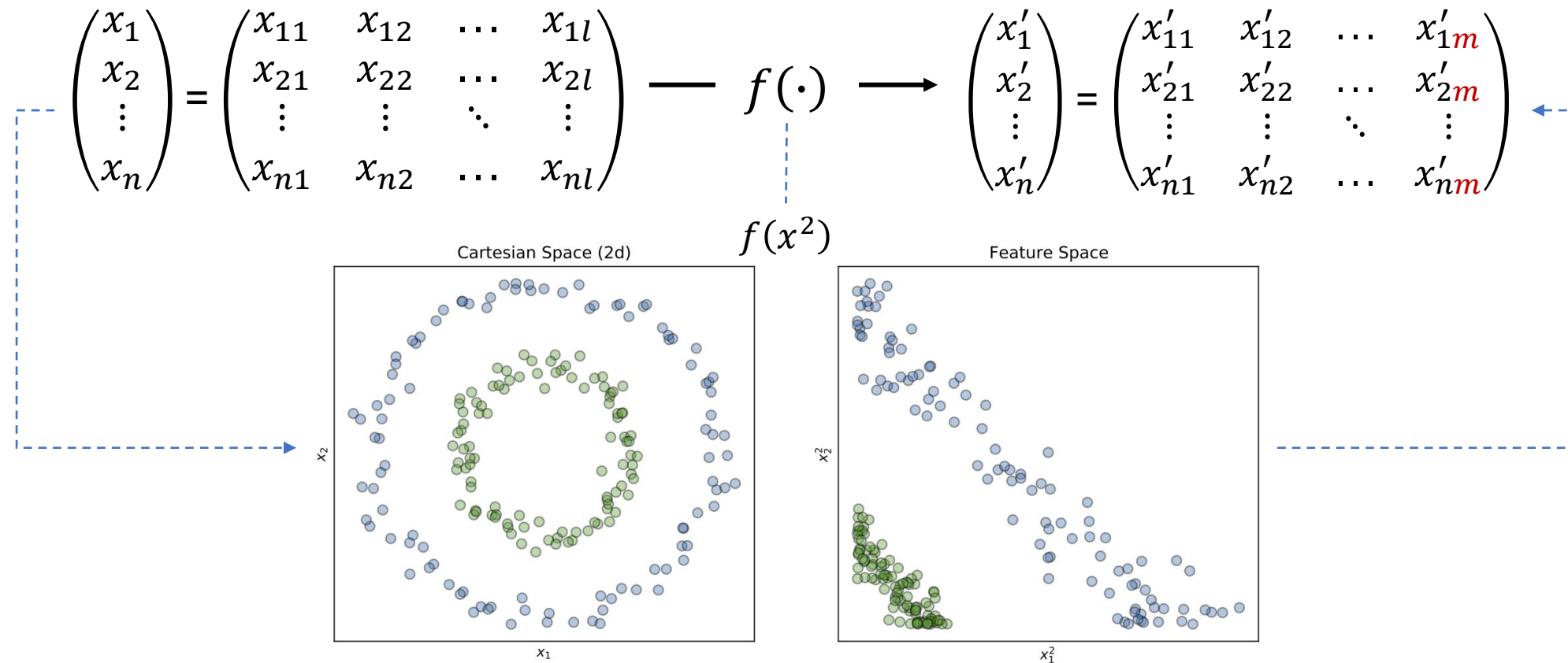
□ Next

- Pattern mining



[Reference] Representation learning

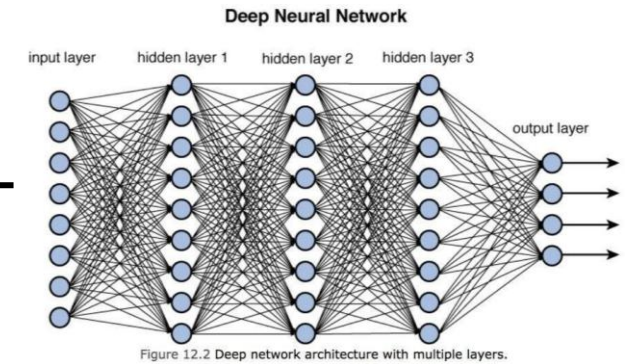
- Learning a transformation function of raw data into a representation that makes it easier to perform downstream tasks
 - i.e., automatic feature engineering



[Reference] Representation learning

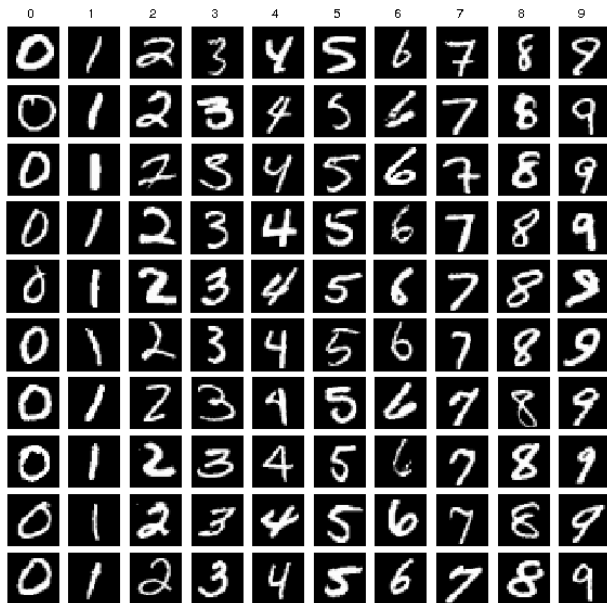
□ Deep representation learning

- Use a deep neural network to learn the transformation function
- Aim to optimize the distances of data objects
 - e.g., contrastive learning

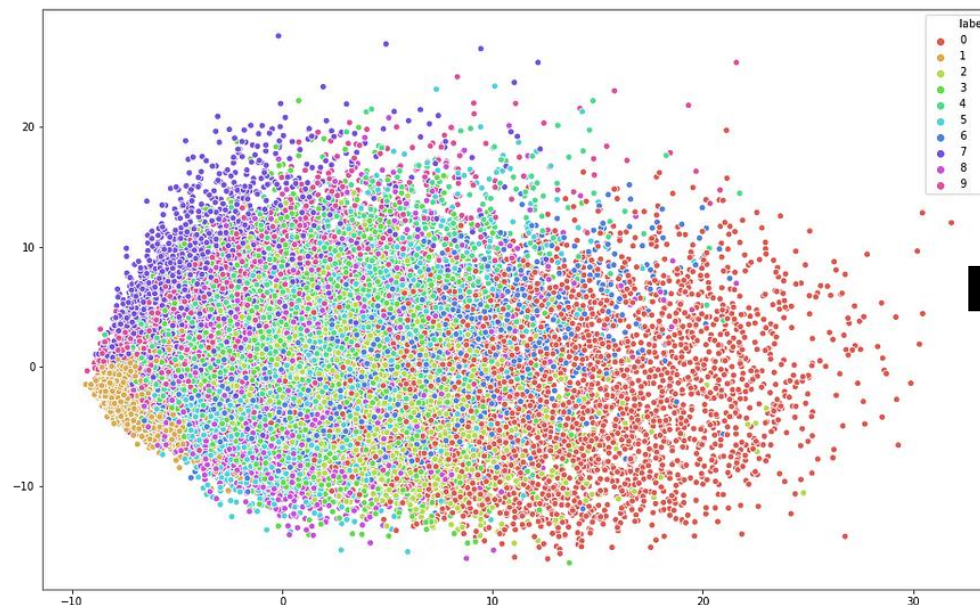


$$d(x_i, x_j) \rightarrow f(\cdot) \rightarrow d(f(x_i), f(x_j))$$

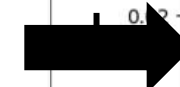
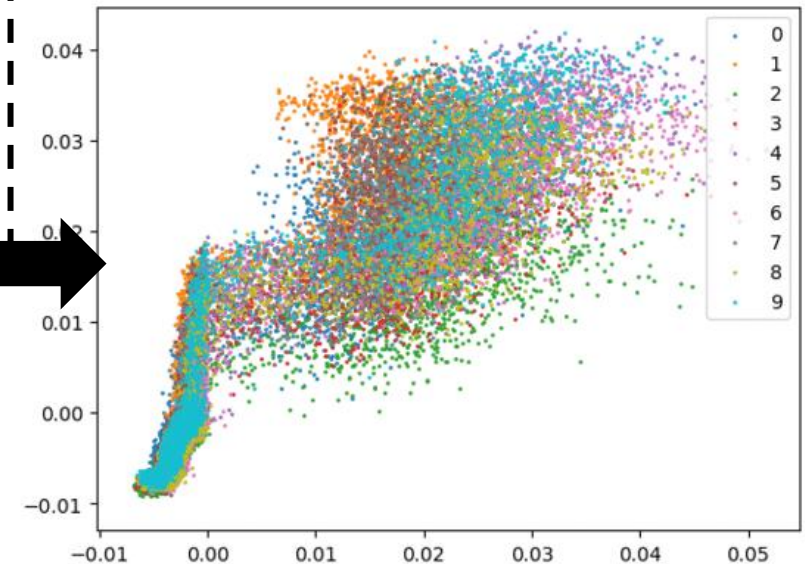
Original image



(Linear) 2D transformation



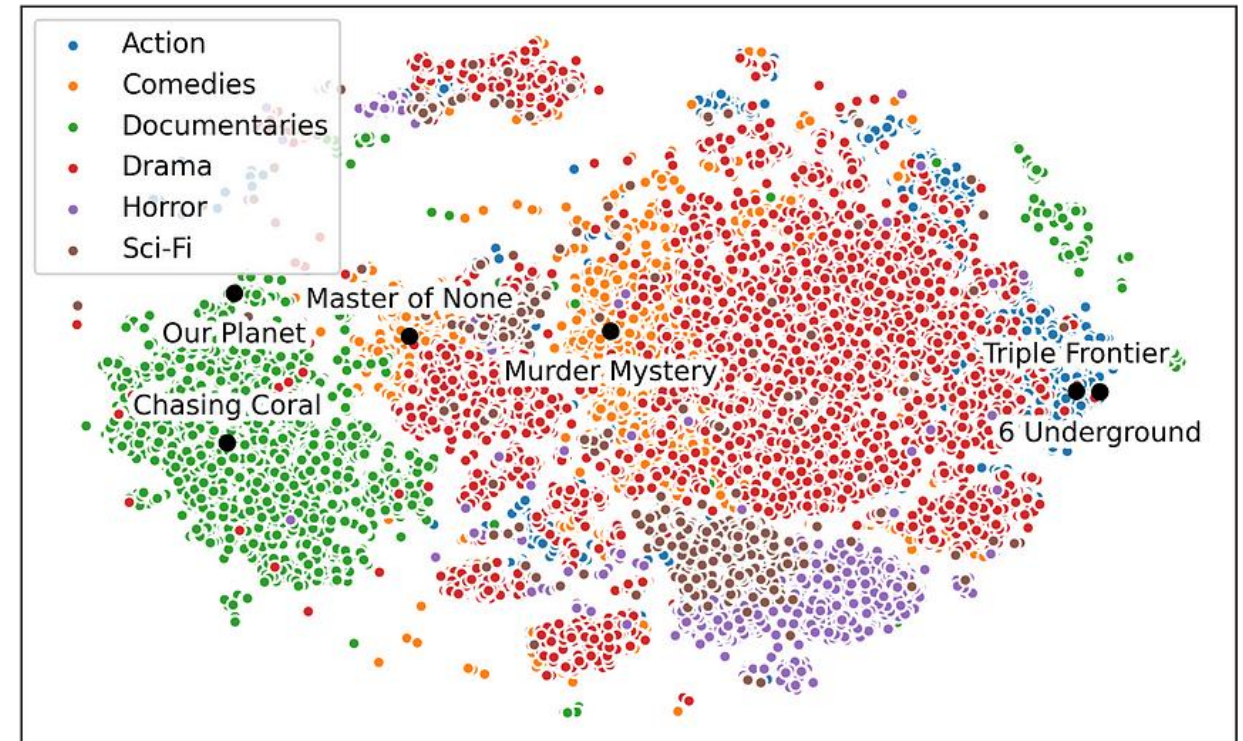
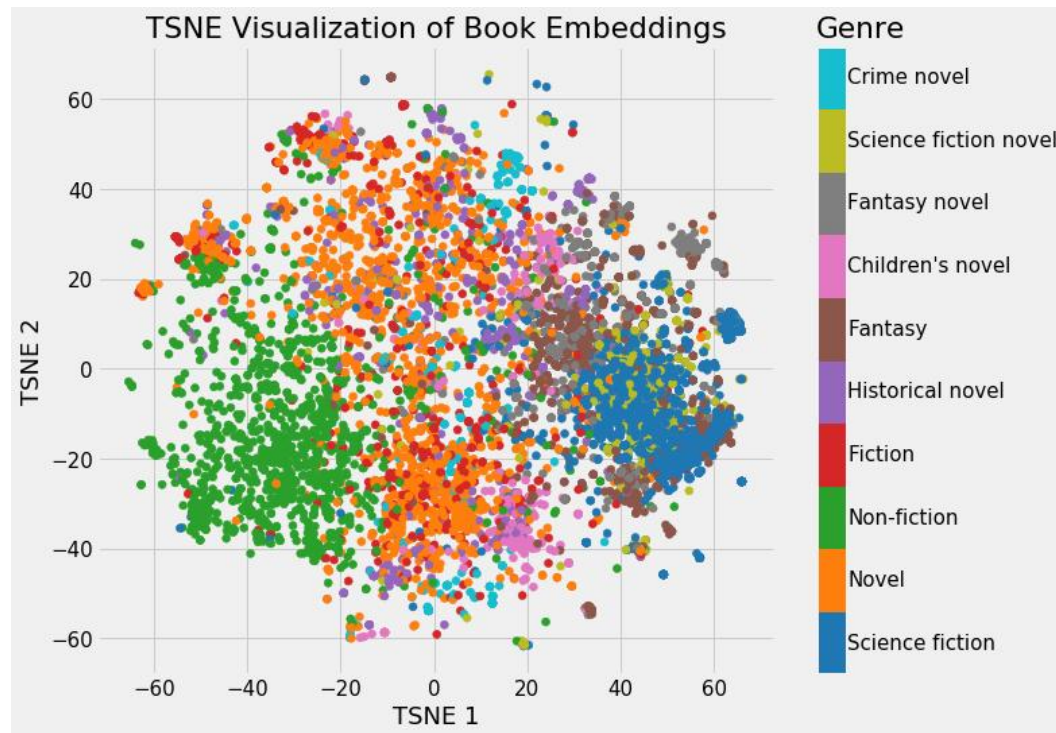
Deep representation learning in 2D



[Reference] Representation learning

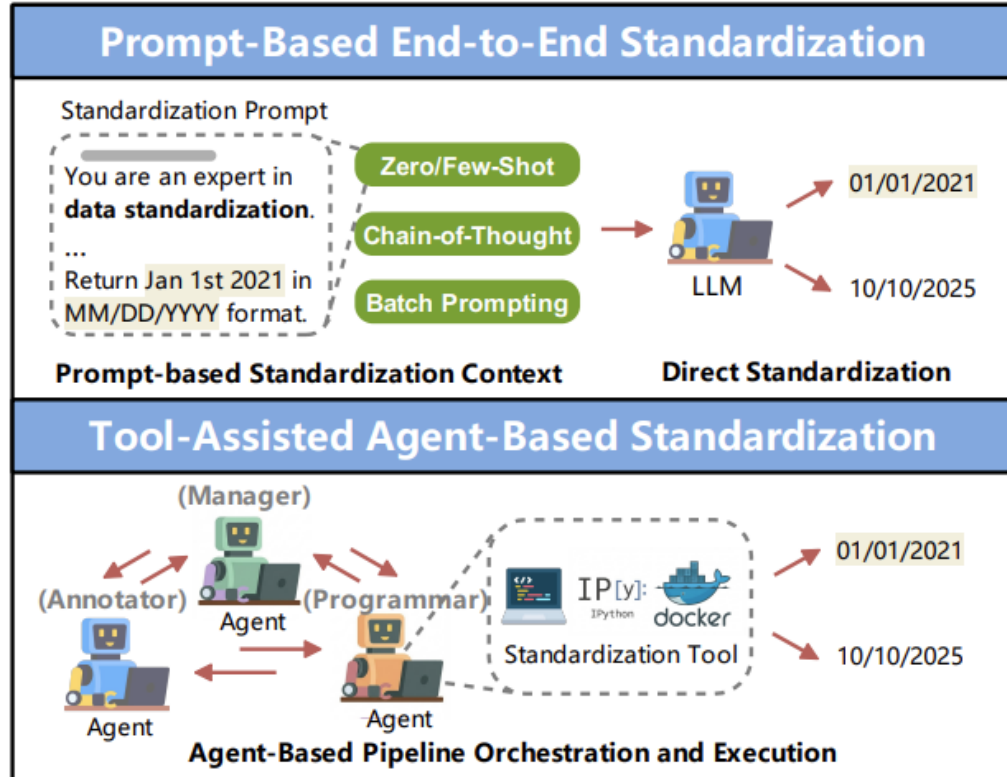
□ Examples

- Novel embedding, Movie embedding, User embedding, Time-series embedding, ...

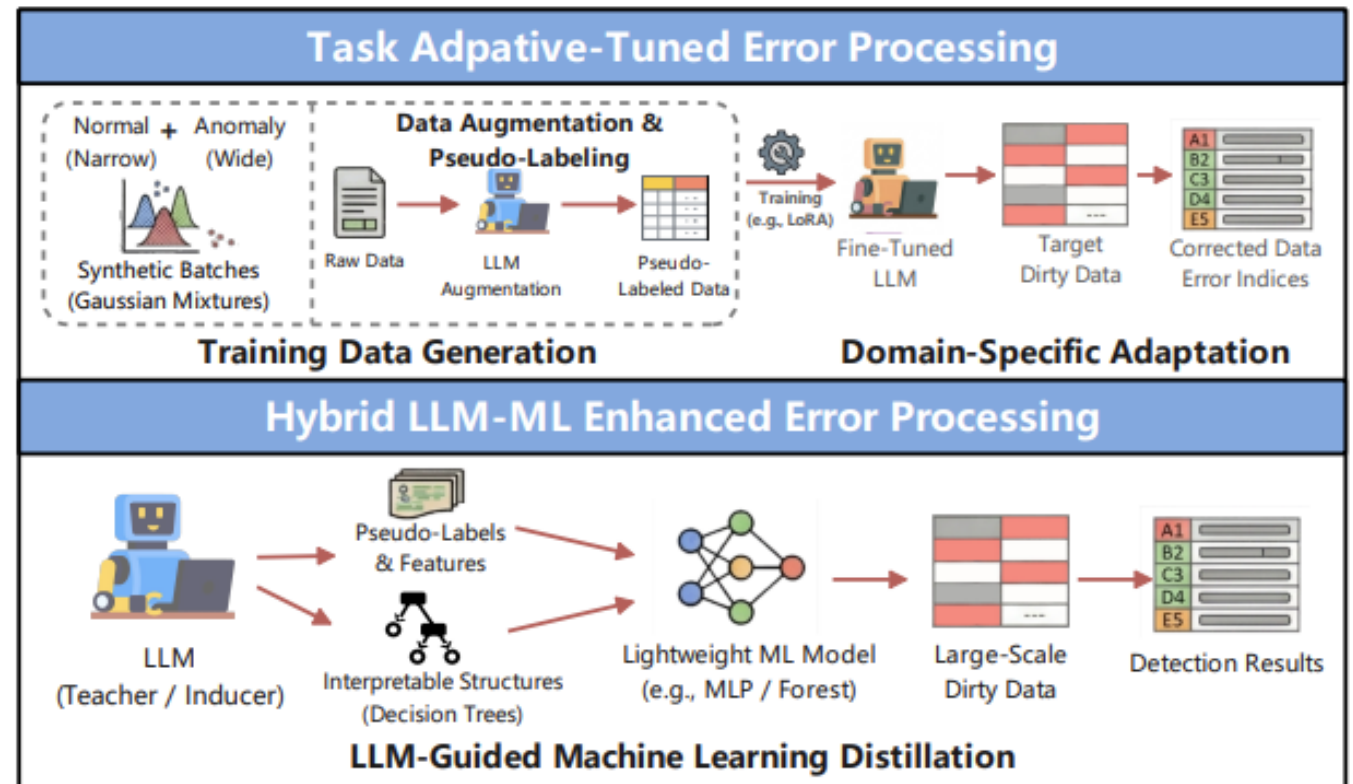


Back

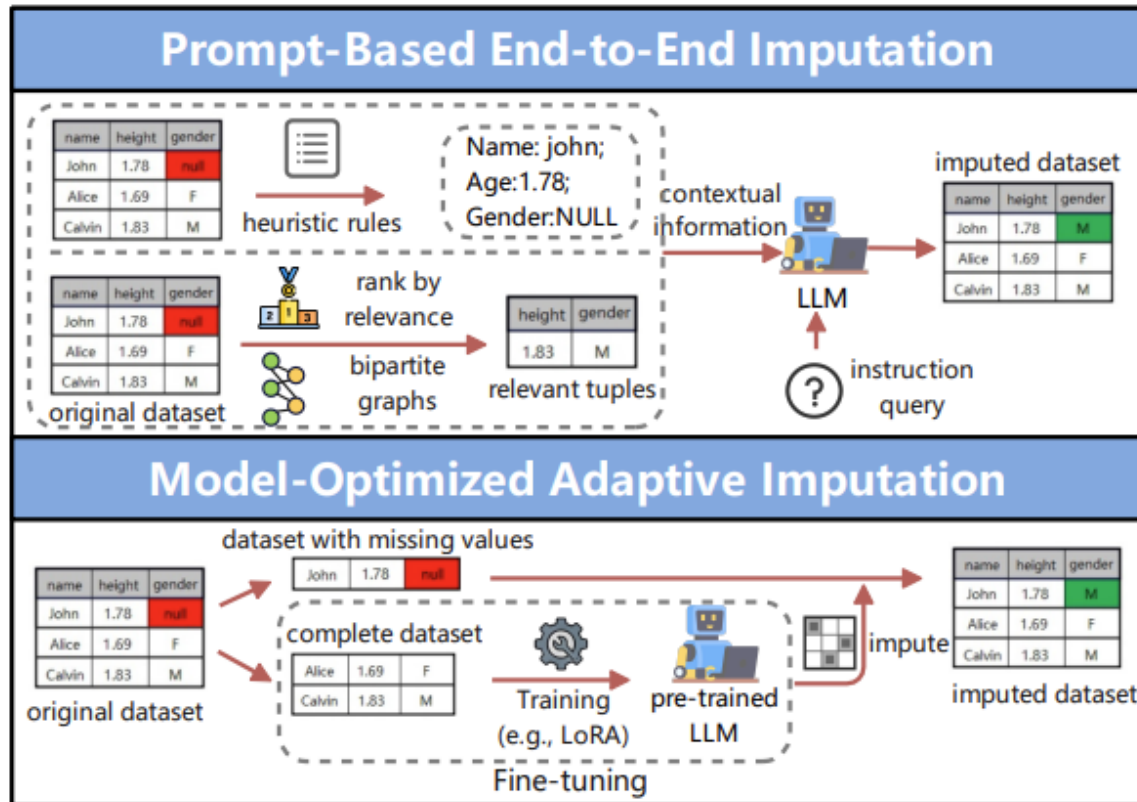
[Reference] Data Preparation by Agent!



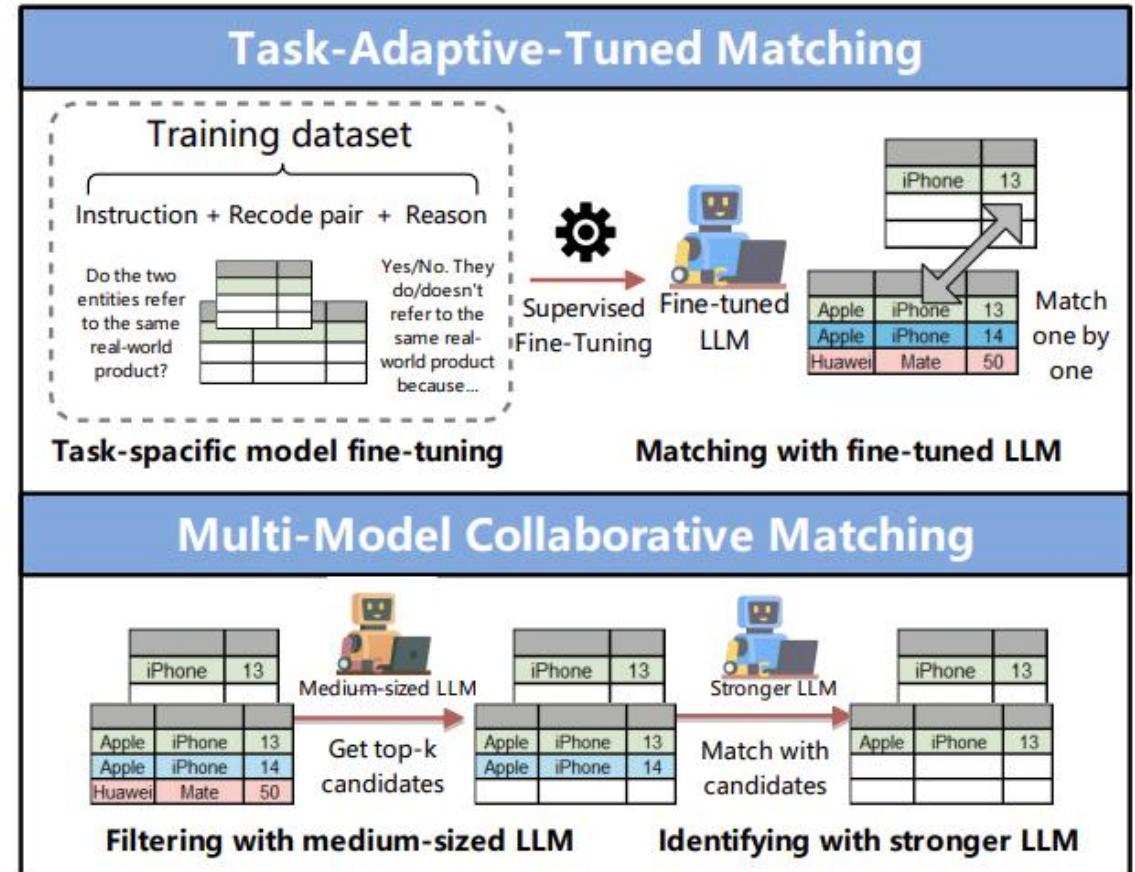
<https://arxiv.org/pdf/2601.17058>



[Reference] Data Preparation by Agent!



<https://arxiv.org/pdf/2601.17058>



[Reference] KDD Cup 2026 (optional topic for term project)

❑ Data Agent Challenge

- **You may pre-form your own team to participate in this competition.** (Note that, by default, team formation for the term project is randomized)
 - All teams must successfully complete Phase 2, in addition to final report and presentation for term project.
 - Credits for cloud/API usage might be supported upon demands/requests
- Official website: <https://dataagent.top/> (in affiliation with KDD 2026: <https://kdd2026.kdd.org/>)
- Schedule
 - Preview | Mar 15, 2026 | Sample Data & Baselines Release
 - **Phase 1 | Apr 1, 2026 | Competition Officially Opens**
 - Phase 1 | Apr 1 – May 15, 2026 | Phase 1: Open Competition
 - **Phase 2 | May 20, 2026 | Phase 2 Begins**
 - Phase 2 | May 31, 2026 | Registration Freeze
 - **Phase 2 | Jun 30, 2026 | Final Submission Deadline**
 - Final | Jul 15, 2026 | Winners Notified
 - Final | Aug 9, 2026 | KDD 2026 Announcement